Validated Dirichlet approximation and weighted arithmetic moments

GRH research working note

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Abstract

Adaptive outward dyadic arithmetic and theta-integral quadrature give a requested-error construction for three fixed completed Dirichlet functions. Fresh samples and executed rational readout training certify the supplied uniform 10^{-24} examples on the stated compact; no practical extreme-precision runtime bound is asserted. Separately, a fixed-twist sextic Hecke-family moment improves an original masked remainder estimate, while retaining a positive cutoff loss. We give the precise external analytic inputs and distinguish these results from zero-location and spectral determinant claims. No proof of GRH or complete Lean formalization is asserted.

1 A finite, validated approximation result

Let $\chi_3 = (0, 1, -1)$, $\chi_4 = (0, 1, 0, -1)$ and $\chi_5 = (0, 1, -1, -1, 1)$, extended periodically, with parities $a = 1, 1, 0$. We approximate the completed functions

$$\Lambda_q(s) = (q/\pi)^{(s+a)/2}\Gamma((s + a)/2)L(s, \chi_q).$$

The primitive theta/Mellin identity is the explicit classical analytic input; no hypothesis about zeros is used. These three examples do not assert a uniform result for all conductors, general Hecke functions or Artin holomorphy.

Theorem 1 (Scope of the supplied finite certificate)**.** *For each $q = 3, 4, 5$, the supplied exact-rational calculation constructs a complex polynomial P_q of degree 28 satisfying*

$$\sup_{|s-1/2|\le 7/10} |P_q(s) - \Lambda_q(s)| < 5 \cdot 10^{-15}.$$

The computation includes enclosures of all 64 true Fourier-circle nodes, completed-function sample error, analytic truncation, Fourier aliasing, the exact closed-form 180-step gradient-descent iterate and coefficient midpoint rounding. The Python implementation and informal analytic proof have independent checks; this is not a kernel-checked Lean theorem.

Proof. Write $F(c+z) = \sum_{n\ge 0} a_n z^n$, with $c = 1/2$, $R = 7/10$, $\rho = 6/5$, $r = 12/5$. The independently recomputed rational theta majorants are strictly below $M_q = 4, 10, 6$, respectively, on $|z| \le r$. Use true roots $\zeta_j = e^{2\pi ij/64}$ and rational labels y_j with proved error $|y_j - F(c + \rho\zeta_j)| \le \eta$. Use the real Euclidean gradient on complex coefficient coordinates for

$$\mathcal{L}(b) = \frac{1}{128}\sum_{j=0}^{63}\left|\sum_{n=0}^{28} b_n\zeta_j^n - y_j\right|^2.$$

Discrete Fourier orthogonality gives the least-squares optimum $b_n^* = 64^{-1}\sum_j y_j\bar{\zeta}_j^n$, $0 \le n \le 28$, and gradient $\nabla\mathcal{L}(b) = b - b^*$. For step $1/4$ and zero initialization, the exact 180-step iterate is

$b_n^{180} = (1-(3/4)^{180})b_n^*$. The code evaluates this closed form; it does not claim an explicit180-update optimizer run.

Set $u = R/\rho = 7/12$, $v = R/r = 7/24$ and $w = (\rho/r)^{64} = 2^{-64}$. Let $\widehat{b}_n$ be an outward rectangle enclosing b_n^* and ϵ_n the sum of the coordinate halfwidths of the rectangle for b_n^{180}. Choose each actual polynomial coefficient as that rectangle's exact rational midpoint c_n. The normalized basis is explicit:

$$P_q(s) = \sum_{n=0}^{28} c_n \left(\frac{s-1/2}{6/5}\right)^n.$$

Thus the exported coefficient with index n multiplies the normalized variable, rather than the raw power $(s-1/2)^n$. Its error on the compact disk is bounded by

$$E_{\text{opt}} = (3/4)^{180} \sum_{n=0}^{28} |\widehat{b}_n|_1 u^n, \qquad E_{\text{arithmetic}} = \sum_{n=0}^{28} \epsilon_n u^n,$$

$$E_{\text{oracle}} = \eta \frac{1-u^{29}}{1-u}, \qquad E_{\text{alias}} = M_q \frac{w}{1-w}\frac{1-v^{29}}{1-v},$$

$$E_{\text{Taylor}} = M_q \frac{v^{29}}{1-v}.$$

Here $|\widehat{b}_n|_1$ is the sum of the maximum absolute real and imaginary coordinates, a safe upper bound for every enclosed complex modulus. Cauchy's estimate $|a_n| \le M_q r^{-n}$ and the Fourier identity $b_n^* = a_n\rho^n + \sum_{\ell\ge 1} a_{n+64\ell}\rho^{n+64\ell}$ for exact labels prove the alias and Taylor bounds. Label perturbations contribute at most η per Fourier coefficient. The remaining two terms follow directly from the exact iterate and coefficient rectangles. Each term is rounded upward to a 180-bit dyadic rational; the reported total is the exact sum of those reported terms. The certified totals below are strictly below the asserted threshold. Detailed sample enclosures are proved in the extended edition and encoded with exact integer endpoints. □

Conductor	Maximum sample radius	Uniform polynomial bound
3	3.026×10^{-17}	1.786×10^{-15}
4	7.206×10^{-17}	4.455×10^{-15}
5	1.404×10^{-16}	2.906×10^{-15}

The displayed values are rounded upward; exact rational records establish the strict inequalities.

2 What this does and does not certify

The final JSON records exact rational centers, interval endpoints, source hashes and all five error terms. Float values are display summaries only. A high-precision comparison is an independent diagnostic, not the reason an interval contains its mathematical target. Arithmetic uses outward integer operations, and invalid float/order inputs are rejected, including through typed caches. Earlier pilots and failed validation probes are retained.

This is a fixed finite tolerance/configuration. Increasing accuracy needs new truncation and precision choices; the fixed $N = 11, T = 128$ interface is not an arbitrary-accuracy oracle. The broader approximation existence theorem is separate. The certificate says nothing about the location of zeros: P_q is not required to be real-rooted or a self-adjoint spectral determinant. The independent arithmetic determinant identity, positivity and original signed short-row/covariance estimates remain open. No GRH proof or complete new Lean4 formalization is claimed.

3 Validated primitive operations

Real intervals are closed dyadic intervals $[l2^{-224}, h2^{-224}]$ with integer endpoints; complex intervals are Cartesian products. Addition is exact at this scale. For multiplication take the minimum/maximum of all four integer endpoint products, divide by 2^{224} and round down/up. For a reciprocal whose interval omits zero, reverse the endpoint order and round $2^{448}/h$ down and $2^{448}/l$ up. Negative denominators and scalar division use the same endpoint rules. Each operation encloses all represented real values by monotonicity; induction proves inclusion for every expression built from them. Rectangular complex multiplication uses these real rules. The sum of coordinate halfwidths bounds the complex distance from its exact rational center. Binary floating arguments are never silently converted.

For $0 < x < 1$, the alternating arctangent sum and the endpoint obtained by adding the next signed term enclose $\arctan x$. Sixty terms at $1/5$ and $1/239$, together with Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$, give the interval for π. The negative coefficient reverses the second interval. The identity follows from tangent addition and the range of the two angles; no decimal value of π is a trusted input.

For rational $x > 0$, range-reduce $x = 2^k v$, $1 \le v < 2$. With $z = (v-1)/(v+1) \le 1/3$,

$$\log v = 2\sum_{j=0}^{J-1} \frac{z^{2j+1}}{2j+1} + \mathcal{R}, \qquad 0 \le \mathcal{R} \le \frac{2z^{2J+1}}{(2J+1)(1-z^2)}.$$

The geometric tail bounds each remaining denominator below by $2J+1$. Use $J = 92$, the analogous expression for $\log 2$, and sign-aware interval multiplication by k. This covers negative k as well.

For complex z, repeatedly halve until its interval ℓ^1 magnitude bound is $A \le 1/2$. The degree 60 Taylor polynomial for e^z then has remainder at most

$$\frac{A^{61}}{61!}\frac{1}{1-A/62}.$$

Successive absolute terms have ratio at most $A/62$, so geometric comparison proves this bound. Add its symmetric interval to each coordinate, then square outward the required number of times. The identity $e^{2z} = (e^z)^2$ and interval inclusion prove the final enclosure. Caches are type-sensitive: otherwise a cached exact argument can accidentally bypass rejection of an equal-valued float. No implementation change is allowed during the recorded run; the producer verifies source bytes again before emitting its certificate.

4 Recomputing the compact growth constants

For the larger growth disk, $c = 1/2$, $r = 12/5$, take $k = 2$ when $a = 1$ and $k = 1$ when $a = 0$. Both real Mellin exponents on that disk are at most k. Define

$$Q_k(x) = \sum_{j=0}^{k} \frac{k!}{(k-j)!x^{j+1}}, \qquad U_{16}(x) = (1+x/16)^{-16}, \qquad \delta = 3/q.$$

Integration by parts gives $\int_1^\infty t^k e^{-xt}\,dt = e^{-x}Q_k(x)$. The same Gaussian split and geometric-tail comparison used below, now with cutoff $n_0 = 8$, imply the exact rational majorant

$$\widehat{M}_q = 2\left[\sum_{n=1}^{7} n^a U_{16}(\delta n^2)Q_k(\delta n^2) + C_a Q_k(64\delta)\frac{U_{16}(32\delta)}{1-U_{16}(8\delta)}\right], \quad C_0 = 1, \quad C_1 = (1+\delta^{-1})/2.$$

Replacing the actual $d = \pi/q$ by the smaller positive δ increases the integral majorant. Exact fraction comparison gives $\widehat{M}_3 < 4$, $\widehat{M}_4 < 10$, $\widehat{M}_5 < 6$. The producer recomputes these fractions, asserts the strict comparisons and exports them; the constants in the readout proof are thus checked rather than supplied by a floating growth estimate.

5 The completed-function integral and its tails

Let $d = \pi/q$ and $\theta_q(t) = \sum_{n\geq 1} \chi_q(n) n^a e^{-dn^2 t}$. The standard primitive theta transformation, followed by splitting the Mellin integral at1, gives

$$\Lambda_q(s) = \int_1^\infty \theta_q(t) t^{(s+a)/2-1}\, dt + \int_1^\infty \theta_q(t) t^{(1-s+a)/2-1}\, dt.$$

The second root-number coefficient is1 for these three characters: their finite Gauss sums are $i\sqrt{3}$, $2i$ and $\sqrt{5}$, respectively. For the last, $2\cos(2\pi/5) - 2\cos(4\pi/5) = \sqrt{5}$; the first two are direct conjugate-root differences. Division by $i^a\sqrt{q}$ gives the claimed normalization. Exponential decay makes the two integrals entire. This classical theta identity, not an assumption of Artin holomorphy or desired positivity, is the analytic input of the implementation.

The actual circle nodes are enclosed by computing $e^{2\pi ij/64}$ with the validated primitives. All input boxes fit $D = [-1, 2] + i[-4/3, 4/3]$. Thus both exponents $p = (s+a)/2-1$ and $p' = (1-s+a)/2-1$ satisfy $|\Re p| \leq 2$, $|\Im p| \leq 2/3$. The domain guard precedes all term discards. Retain $1 \leq n \leq 11$ and integrate $1 \leq t \leq 128$.

Put $\delta = 3/q < d$, $U(x) = (1+x/128)^{-128}$ and $Q(x) = x^{-1} + 2x^{-2} + 2x^{-3}$. Since $(1+x/128)^{128} \leq e^x$, $e^{-x} \leq U(x) < 1$ for $x > 0$. Integration by parts gives $\int_1^\infty t^2 e^{-xt}\, dt = e^{-x}Q(x)$. With $C_0 = 1$, $C_1 = (1+\delta^{-1})/2$ the omitted-n tail of both integrals is

$$2C_a Q(\delta 12^2) \frac{U(\delta 12^2/2)}{1 - U(12\delta)}.$$

Indeed $n^a e^{-\delta n^2/2} \leq C_a$, the function Q decreases, and $(12+j)^2 \geq 12^2 + 24j$. These inequalities yield a geometric bound after splitting the exponent in halves. The retained-n time tail is at most

$$\sum_{n=1}^{11} 2n^a 128^3 U(128\delta n^2) Q(128\delta n^2).$$

It follows by substituting $t = 128u$ and majorizing $t^{\Re p}$ by t^2. The two tails are disjoint: omitted n use the full integral, whereas retained n contribute only $t > 128$.

6 High-order integration with a certified remainder

Split each dyadic interval $[2^j, 2^{j+1}]$, $j = 0, \ldots, 6$, into four equal panels. For its midpoint m and halfwidth h, $h/m \leq 1/9$. For $f(z) = e^{-\lambda z} z^p$, $\lambda = \pi n^2/q$, the principal logarithm is analytic on $|z - m| \leq \rho = m/2$. On that circle,

$$|f(z)| \leq M_\rho = 3\max\{(3m/2)^2, (m/2)^{-2}, 1\} U(\delta n^2 m/2).$$

Here $\Re z \geq m/2$, $|\arg z| \leq 1$ and $e^{|\Im p|} \leq e^{2/3} \leq 3$. The maximum handles both signs of $\Re p$. Cauchy's coefficient estimate and cancellation of odd powers on $[-h, h]$ therefore bound the error after degree $2K$ by

$$\frac{2hM_\rho (h/\rho)^{2K+2}}{(2K+3)(1-(h/\rho)^2)}.$$

Indeed integrate each remaining even term, replace its denominator by the first omitted denominator, then sum a geometric series. Here $K = 12$. Multiply by $2n^a$ for both Mellin terms and sum all 28 panels and retained n. Keeping this majorant even for later discarded terms only overcounts error.

To evaluate the polynomial stably, set $u = h/m$, $q_{-1} = 0$, $q_0 = 1$ and use

$$q_{j+1} = \frac{u(p - \lambda m - j)q_j - (\lambda m)u^2 q_{j-1}}{j+1}.$$

Multiplying $f' = (-\lambda + p/t)f$ by $t = m + x$ and matching coefficients proves this recurrence, with $q_j = f^{(j)}(m)h^j/(j!f(m))$. The integrated polynomial is $2he^{-\lambda m}m^p \sum_{j=0}^{K} q_{2j}/(2j+1)$; all its arithmetic is outward.

Fixed absolute exponential-rounding errors can be amplified by high-order coefficients in late panels. The code therefore first bounds a complete (n, panel) pair, including both powers, by

$$4hn^a(m+h)^2 U(\delta n^2(m-h)).$$

For real $t \geq 1$ its phase has modulus1 and $t^{\Re p} \leq (m+h)^2$, proving the bound. If it is below 2^{-90}, skip that pair entirely and add its bound to the final radius. This is controlled truncation, not numerical underflow. The final sample radius is the interval-sum halfwidth plus the omitted-n, time, Cauchy and discard bounds. All are recorded separately and checked as nonnegative. No high-precision special-function routine is trusted here.

7 Reproduction and remaining interfaces

Run the preserved source on the existing Linux Python environment with no new packages: `python3` on `examples/analytic_oracle/theta_oracle.py`, with argument `-output NEW`. It refuses to overwrite an existing experiment directory. The final-v5 experiment contains 192 samples, three rational degree 28 polynomials, copied source and SHA-256 hashes. Independent validation checks rational endpoint arithmetic, transcendental enclosures, 1160 exact coefficient identities, high- precision completed-function comparisons and the exported component sums.

The fixed-parameter oracle solves a concrete evaluation/certification gap. It does not identify a positive arithmetic operator, force real-rooted polynomials, prove the signed original-family short-row bound or supply complete Lean4 verification. Surface/order and neural architectures may use these certificates as independent targets, but a small learned loss is not a substitute for the remaining arithmetic and categorical theorems.

8 Requested-error readouts from preserved certified labels

The accepted final-v5 sample certificate is preserved unchanged. For each $q = 3, 4, 5$, it supplies $m = 64$ rational complex labels y_j, bounds $|y_j - \Lambda_q(c + \rho\zeta_j)| \leq \eta_j$, and outward rectangles enclosing the true roots $\zeta_j = e^{2\pi ij/m}$, where $c = 1/2$, $\rho = 6/5$. The classical primitive theta/Mellin identity and accepted growth bounds $M_q = 4, 10, 6$ on $|s - c| \leq r = 12/5$ remain the analytic inputs. This extension accepts an exact rational error request $\varepsilon > 0$ on the fixed compact $|s - c| \leq R = 7/10$. It changes the finite readout, while retaining the certified sample errors and the 64-node alias budget. It is not an arbitrary-accuracy theta-sampling interface.

Theorem 2 (Executed rational readout with explicit refusal)**.** *For each supported conductor and rational request, the finite-data selection below either returns a normalized rational polynomial P*

and an executed, exactly replayable gradient-descent trace with

$$\sup_{|s-c|\le R} |P(s) - \Lambda_q(s)| \le \varepsilon,$$

or refuses the request with the exact minimum base budget over all 64 *allowed degrees. Refusal concerns attainability by this particular fixed-data majorant; it is not an impossibility theorem for approximation.*

Proof. Let B_n be an outward rectangle enclosing $b_n^* = m^{-1}\sum_j y_j \bar{\zeta}_j^n$, $0 \le n < 64$, computed from the preserved root rectangles. Put $t_n = \mathrm{mid}(B_n)$ and let e_n be the sum of its coordinate halfwidths. These are exact rationals, and $|t_n - b_n^*| \le e_n$. For a selected $d < 64$ execute, using exact rational operations,

$$b_n^0 = 0, \qquad b_n^{k+1} = \tfrac{3}{4} b_n^k + \tfrac{1}{4} t_n, \qquad 0 \le n \le d.$$

This is genuine gradient descent with step $1/4$ for the explicitly specified real Euclidean objective $J(b) = \frac{1}{2}\sum_{n=0}^{d} |b_n - t_n|^2$, whose gradient is $b - t$. All iterates and exact objective values are retained. The output is literally

$$P_k(s) = \sum_{n=0}^{d} b_n^k \left(\frac{s-c}{\rho}\right)^n.$$

Set $u = R/\rho = 7/12$, $v = R/r = 7/24$, $w = (\rho/r)^{64} = 2^{-64}$, $S_d(x) = \sum_{n=0}^{d} x^n$ and $\eta = \max_j \eta_j$. The five budgets are

$$E_{\mathrm{opt}}(k) = \sum_{n=0}^{d} |b_n^k - t_n|_1 u^n, \qquad E_{\mathrm{oracle}} = \eta S_d(u),$$

$$E_{\mathrm{alias}} = M_q \frac{w}{1-w} S_d(v), \qquad E_{\mathrm{Taylor}} = M_q \frac{v^{d+1}}{1-v},$$

$$E_{\mathrm{arithmetic}} = \sum_{n=0}^{d} e_n u^n.$$

Here $|x + iy|_1 = |x| + |y|$. Cauchy's estimate and true-root Fourier orthogonality give the alias and Taylor terms; certified label errors give the oracle term. Triangle inequality applied to $b_n^k - b_n^* = (b_n^k - t_n) + (t_n - b_n^*)$ gives the other two. Thus their exact sum bounds the compact-uniform error.

Define $C_d = E_{\mathrm{oracle}} + E_{\mathrm{alias}} + E_{\mathrm{Taylor}} + E_{\mathrm{arithmetic}}$. Select the smallest $d < 64$ with $C_d < \varepsilon$; if none exists, report $\min_{d<64} C_d$ and refuse. For a selected degree, $E_{\mathrm{opt}}(k) = (3/4)^k \sum_n |t_n|_1 u^n \longrightarrow 0$. Therefore executing updates until $E_{\mathrm{opt}}(k) \le \varepsilon - C_d$ terminates after finitely many steps. The serialized coefficients, budgets, targets and trace use exact rational endpoints; displayed decimal approximations are not proof inputs. □

9 Objective semantics, precision floors and replay

The original true-circle sample-label least-squares objective is

$$\mathcal{L}_y(b) = \frac{1}{2m} \sum_{j=0}^{m-1} \left| \sum_{n=0}^{d} b_n \zeta_j^n - y_j \right|^2.$$

With $d < m$, Fourier orthogonality gives $A^*A = mI$ and the real Euclidean coordinate gradient $\nabla \mathcal{L}_y(b) = b - b^*$. The executed objective J instead has the exact rational midpoint target t. The

difference $\nabla\mathcal{L}_y - \nabla J = t - b^*$ is enclosed coefficientwise by e_n. Consequently this is an actual rational GD run for J and a certified perturbed-gradient implementation for $\mathcal{L}_y$. It is not the exact optimizer trajectory of the ideal-root objective, nor a replay of neural or quantum training. An interval enclosure of an ideal-root GD trajectory would be a different explicitly labelled object.

Each actual update contracts $b^k - t$ by $3/4$, so the exact stored losses satisfy $J(b^{k+1}) = (9/16)J(b^k)$. A replay recomputes every rational coordinate update, loss and weighted residual. The closed form $b^k = (1 - (3/4)^k)t$ is an independent check on the executed trace, not its producer. Exact rational loop coordinates introduce no further midpoint or floating export error. The arithmetic budget covers the difference between the certified Fourier rectangles and their midpoint targets, including the inherited root enclosure and outward interval operations. A subsequent float conversion requires its own exact conversion-error bound.

The normalized polynomial convention is essential: a stored coefficient with index n multiplies $((s - 1/2)/(6/5))^n$, not $(s - 1/2)^n$. The input certificate is pinned by SHA-256, including its source hashes, all 192 sample enclosures, root rectangles and analytic radius components. Producing source is captured at entry, checked unchanged before emission, and copied into a fresh output directory. An existing output is never overwritten. An independent replay should also check the fixed input hash, all dimensions, degree choice, the first stopping iterate, all five exact terms and their literal sum against the requested rational tolerance.

The attainable boundary of this majorant is

$$C_{\min,q} = \min_{0 \le d < 64} C_{q,d}.$$

Every request $\varepsilon > C_{\min,q}$ has strict budget slack for some degree and the proved finite stop rule. At or below that boundary this interface refuses. The degree minimizing C_d need not be 63: Taylor error decreases with degree while the oracle and arithmetic sums increase. In particular $C_d \ge \eta$ and $C_d \ge M_q 2^{-64}/(1 - 2^{-64})$. Increasing the number of optimization steps does not lower these fixed-data budgets. Approximate boundary values on the stated compact are $7.2929 \cdot 10^{-17}$, $1.7370 \cdot 10^{-16}$ and $3.3741 \cdot 10^{-16}$ for $q = 3, 4, 5$; only the exact exported rational minima are used for acceptance. These are certificate floors, not lower bounds on the actual errors.

A full adaptive theta oracle would separately need proof-preserving choices of omitted-n cutoff, time cutoff, quadrature order, whole-panel discard budget, primitive series lengths, interval precision and serialized-bound precision. The present $N = 11$, $T = 128$, 224-bit arithmetic, 60/92/60 primitive series lengths, 2^{-90} discard threshold and 180-bit rounded budgets cannot support every positive request. Increasing a nominal bit count alone leaves the fixed primitive remainders and analytic tails in place. The preserved labels are not recomputed here. The classical identity, finite arithmetic and informal proof are not new Lean declarations. No zero-location, all-conductor, positive arithmetic determinant, quantum-advantage or GRH conclusion follows.

The interactive replay is available at the certified readout laboratory. All decimal graphics are illustrative views of the linked exact records.

10 Requested-error theta samples for three fixed conductors

For the primitive characters $\chi_3 = (0, 1, -1)$, $\chi_4 = (0, 1, 0, -1)$, $\chi_5 = (0, 1, -1, -1, 1)$ with parities $a = 1, 1, 0$, retain the classical primitive theta/Mellin identity and root number one already stated above. Put $D = [-1, 2] + i[-4/3, 4/3]$ and $\delta = 3/q$. The following adaptive extension is implemented with independent interval and sample checks; it is an informal analytic theorem, not a Lean result.

Theorem 3 (Finite adaptive sample specification)**.** *For every supported conductor, exact rational $s \in D$, and rational $\eta > 0$, the choices below terminate and give a rational complex midpoint y and exact rational radius $R \le \eta$ with $|y - \Lambda_q(s)| \le R$. The same conclusion holds for a specified true circle node when its root and node enclosures are refined on every precision retry.*

Proof. For $x \ge 0$ define the exact rational bound $E(x) = 2^{-\lfloor x \rfloor}$. Since $e > 2$, $e^{-x} \le E(x)$; this bound is nonincreasing and tends to zero. Let $Q(x) = x^{-1} + 2x^{-2} + 2x^{-3}$ and $C_0 = 1$, $C_1 = (1 + \delta^{-1})/2$. For $N \ge 1$, $n_0 = N + 1$, $J \ge 1$, $T = 2^J$, use

$$B_n = 2C_a Q(\delta n_0^2) \frac{E(\delta n_0^2/2)}{1 - E(\delta n_0)},$$

$$B_t = \sum_{n=1}^{N} 2n^a T^3 E(\delta n^2 T) Q(\delta n^2 T).$$

Here $\delta n_0 \ge 6/5$ ensures a positive denominator; $E(x)$ equals one for $0 \le x < 1$, so this safeguard is essential. On the four panels of each $[2^j, 2^{j+1}]$, write midpoint m, halfwidth h, $\rho = m/2$, $v = h/\rho \le 2/9$, and $B(m) = \max\{(m + \rho)^2, (m - \rho)^{-2}, 1\}$. Use the Cauchy remainder

$$B_Q = \sum_{\text{panels}} \sum_{n=1}^{N} \frac{2n^a \, 2h \, 3B(m) E(\delta n^2 m/2) v^{2K+2}}{(2K + 3)(1 - v^2)}.$$

Choose in turn the first $N \ge 1$, first $J \ge 1$, and first $K \ge 0$ for which their respective budgets are at most $\eta/8$. Gaussian decay, then exponential time decay, then $v \le 2/9$ prove termination. These are the accepted tail and Cauchy arguments with each true exponential separately replaced by its upper bound E.

There are $P = 4JN$ potential pairs. A nonzero-character pair may be omitted only if its entire two-power contribution has bound

$$b = 4hn^a (m + h)^2 E(\delta n^2 (m - h)) \le \eta/(8P).$$

Add every omitted bound to B_{discard}, giving $B_{\text{discard}} \le \eta/8$. Evaluate all other integrated Taylor polynomials by the stated exact coefficient recurrence using outward rectangles. Refine both interval precision and all primitive series remainders until the final rectangle's sum of coordinate halfwidths satisfies $R_{\text{arith}} \le \eta/4$. For this fixed finite expression all input and primitive intervals shrink; continuity and outward arithmetic imply $R_{\text{arith}} \to 0$. Precision doubling therefore terminates, and the sum of the five raw components is at most $3\eta/4$. Choose an integer output scale B such that $5\,2^{-B} \le \eta/8$ and round each component upward by $\text{up}_B(x) = 2^{-B}\lceil 2^B x \rceil$. The reported radius is their literal exact sum. With $\mathbf{b} = (R_{\text{arith}}, B_n, B_t, B_Q, B_{\text{discard}})$, it obeys

$$|y - \Lambda_q(s)| \le \sum_{i=1}^{5} \text{up}_B(b_i) \le \frac{3\eta}{4} + 5\,2^{-B} \le \frac{7\eta}{8} < \eta.$$

The output retains exact rational midpoints, rounded components, B, raw interval radius, parameters, input metadata and source hashes. The raw analytic bounds are exactly reconstructible; they need not be serialized as potentially enormous fractions. The output precision varies with the request and introduces no fixed reporting floor. □

This fixed-conductor sample result uses no hypothesis on zeros. It does not remove a downstream fixed-node Fourier alias floor, supply an all-conductor theorem or a positive arithmetic determinant, or prove GRH.

11 Adaptive sampling parameters and precision termination

The primitive theta/Mellin identity with root number one yields

$$\Lambda_q(s) = \sum_{n\geq 1} \chi_q(n) n^a \int_1^\infty e^{-\pi n^2 t/q} \big(t^{(s+a)/2-1} + t^{(1-s+a)/2-1}\big)\, dt.$$

On $D = [-1,2] + i[-4/3, 4/3]$, both exponents p satisfy $|\Re p| \leq 2$ and $|\Im p| \leq 2/3$. For exact $x \geq 0$, put $E(x) = 2^{-\lfloor x \rfloor}$. The elementary inequality $e > 2$ gives $e^{-x} \leq e^{-\lfloor x \rfloor} \leq E(x)$ and $E(x) \leq 2e^{-x\log 2}$. Use this solely as a pointwise nonincreasing upper bound; it is not multiplicative and has no asserted exponential integral identity.

Integration by parts gives $\int_1^\infty t^2 e^{-xt}\, dt = e^{-x}Q(x)$ for $Q(x) = x^{-1} + 2x^{-2} + 2x^{-3}$. For $a = 0$ the bound $n^a e^{-\delta n^2/2} \leq C_0 = 1$ is immediate. For $a = 1$, $n \leq (1+n^2)/2$ and $n^2 e^{-\delta n^2/2} \leq 2/(e\delta) \leq 1/\delta$ give $C_1 = (1+1/\delta)/2$. For $n = n_0 + j$, the inequality $n^2 \geq n_0^2 + 2n_0 j$ therefore bounds the omitted-n sum by

$$2C_a Q(\delta n_0^2) \frac{e^{-\delta n_0^2/2}}{1 - e^{-\delta n_0}} \leq 2C_a Q(\delta n_0^2) \frac{E(\delta n_0^2/2)}{1 - E(\delta n_0)} = B_n.$$

Take $n_0 = N + 1 \geq 2$; then $\delta n_0 \geq 6/5$ and the latter denominator is positive. Substituting $t = Tu$ in each retained-n time tail gives

$$B_t = \sum_{n=1}^{N} 2n^a T^3 E(\delta n^2 T) Q(\delta n^2 T).$$

The two tails are disjoint because the omitted-n tail uses the whole integral and the time tail uses only retained n.

For $j = 0, \ldots, J-1$ and $\ell = 0, \ldots, 3$ take

$$m = 2^j \left(1 + \frac{2\ell+1}{8}\right), \qquad h = \frac{2^j}{8}, \qquad T = 2^J.$$

These panels cover $[1, T]$ and have $h/m \leq 1/9$. On $|z - m| = \rho = m/2$, principal log is analytic, $\Re z \geq m/2$, and $|\arg z| \leq 1$. For $f(z) = e^{-\pi n^2 z/q} z^p$, the angular multiplier is at most $e^{2/3} \leq 3$, whence

$$|f(z)| \leq 3B(m)E(\delta n^2 m/2), \qquad B(m) = \max\{(3m/2)^2, (m/2)^{-2}, 1\}.$$

Cauchy's coefficient estimate, cancellation of odd terms, and geometric summation yield the B_Q in Theorem 3. Including zero-character or later discarded terms in that sum is harmless. Gaussian, time, and geometric decay prove termination of the sequential first-success searches with $B_n, B_t, B_Q \leq \eta/8$. For fixed q and $L = \max\{1, \log(1/\eta)\}$ their asymptotic scales can be chosen as $N = O(\sqrt{L})$, $T = O(L)$, $J = O(\log L)$, and $K = O(L)$. Polynomial prefactors in N, T, J contribute only $O(\log L)$ to the logarithm of the error estimates; this is a truncation-size statement, not a rational-arithmetic bit-complexity bound.

For real $t \in [m-h, m+h]$, $t \geq 1$ and $t^{\Re p} \leq (m+h)^2$. The bound for a whole pair, including both powers, is therefore $4hn^a(m+h)^2 E(\delta n^2(m-h))$. There are $P = 4JN$ potential pairs; summing only bounds at most $\eta/(8P)$ gives $B_{\text{discard}} \leq \eta/8$. No partial recurrence or isolated exponential is discarded. For active pairs retain the coefficient recurrence

$$r_{-1} = 0, \quad r_0 = 1, \quad r_{j+1} = \frac{(h/m)(p - \lambda m - j) r_j - \lambda m (h/m)^2 r_{j-1}}{j+1}, \qquad \lambda = \pi n^2/q.$$

Coefficient comparison in $tf' = (-\lambda t + p)f$ proves that the integrated Taylor polynomial is

$$2he^{p\log m - \lambda m} \sum_{j=0}^{K} \frac{r_{2j}}{2j+1}.$$

Outward arithmetic encloses its exact value, including the signed character weight and both powers.

Use an isolated p-bit dyadic context, with all primitive remainders refined as p increases. Put $\delta_p = 2^{-p-16}$. Each arctangent series separately stops when its next-term bound is at most δ_p, giving total weighted analytic tail width at most $20\delta_p$ in Machin's formula, before outward dyadic rounding. Each reduced logarithm series separately has tail at most δ_p; for $x = 2^k v$, the combined bound is $(1 + |k|)\delta_p$, before rounding. For the complex exponential, after reduction to $A \le 1/2$, choose the Taylor order so the bound

$$\frac{A^{H+1}}{(H+1)!} \frac{1}{1 - A/(H+2)}.$$

is at most δ_p. Choose H by finite exact search; its denominator is positive. Add a symmetric exponential error enclosure to each coordinate before outward squaring. The fixed guard16 is not itself an acceptance bound: the final complete rectangle radius must satisfy $R_{\text{arith}} \le \eta/4$. All interval caches must include p and preserve exact input-type guards. The successful p and retry precisions/radii are recorded. Individual series lengths are determined by these, the exact inputs and the pinned source and are reconstructible by replay; they are not explicitly stored.

For fixed N,J,K and exact target point, the expression has finitely many operations and only nonzero fixed integer denominators. Each primitive input is bounded for sufficiently large p; exponential halving counts are consequently bounded. Exact primitive tails and dyadic rounding widths tend to zero. Induction through the finite continuous expression, including finitely many outward squarings, gives $R_{\text{arith}} \to 0$. Doubling p after a failed comparison terminates. The argument also covers true roots and nodes regenerated at every p, since their widths tend to zero. A fixed nondegenerate s rectangle has no corresponding shrinking-input claim.

A growing-order primitive estimate. To justify convergence despite a growing exponential Taylor order, write $\epsilon_p = 2^{-p}$, $\delta_p = 2^{-p-16}$, and let $d(X)$ be the sum of a rectangle's coordinate widths and $m(X)$ its ℓ^1 magnitude bound. The rounded reduced input W satisfies $a = m(W) \le 1/2 + 2\epsilon_p < 3/5$ and $d(W) \le 2^{-k} d(Z) + 4\epsilon_p$; the remainder is at most 2^{-H}, so the first accepted order satisfies $H \le p + 18$. For the computed term boxes, outward multiplication and integer division give $m_j \le a m_{j-1} + 6\epsilon_p$ and $d_j \le a d_{j-1} + m_{j-1} d(W) + 12\epsilon_p$. Thus $m_j \le a^j + 15\epsilon_p$ and $\sum_{j=0}^{H} d_j \le (25/4) d(W) + 75H\epsilon_p$; adding the remainder increases width by at most $4(\delta_p + \epsilon_p)$. The pre-squaring width is therefore $O(d(Z) + p2^{-p})$ and its magnitude is bounded (indeed below 3 for $p \ge 32$). Each outward square has width at most $2m(X)d(X) + 8\epsilon_p$; the bounded number of squarings consequently preserves convergence to zero. Finally, exact rational partial sums and endpoint rounding give Machin width at most $20\delta_p + 40\epsilon_p$, and logarithm width at most $(1 + |k|)(\delta_p + 2\epsilon_p)$ for fixed $x = 2^k v$. These estimates justify the primitive-width step without assuming a fixed Taylor order or asserting a global arithmetic complexity bound.

For serialization, initialize

$$B = \max\{32, \text{bitlen}(\text{den}\,\eta) - \text{bitlen}(\text{num}\,\eta) + 8\}$$

and increase B until $5\,2^{-B} \le \eta/8$. Write $\widehat{x} = 2^{-B}\lceil 2^B x \rceil$ for each of the five raw components. Their exact reported sum is at most $3\eta/4 + 5\,2^{-B} \le 7\eta/8$, and every component remains an upper bound. The output records B, the raw interval radius and complete rectangle, N/J/K, discard threshold/pair

counts, exact target metadata and source hashes. These reconstruct all raw analytic bounds and pair sets without serializing huge exact tail denominators. For the practical requests below B is87 and114, respectively. Exact endpoints remain rational; the mathematical specification assumes an integer representation sufficient for the requested size. Dynamic rounding removes the old fixed reporting floor. For rational s, validate its exact coordinates in closed D before conversion: an outward dyadic enclosure of $\Im s = 4/3$ extends beyond D, but the analytic bounds concern the true point. Circle nodes satisfy the strict exact bounds $-7/10 \le \Re s \le 17/10$, $|\Im s| \le 6/5$.

The exact parameter-design checker gives

η	q	N	J	T	K	active pairs
10^{-24}	3	12	7	128	19	71
10^{-24}	4	14	8	256	20	65
10^{-24}	5	16	8	256	20	111
10^{-32}	3	14	7	128	25	84
10^{-32}	4	16	8	256	26	72
10^{-32}	5	18	8	256	26	129

with exact rational budgets and no special-function oracle. The arithmetic precision is selected from the actual interval radius, not inferred from this table. This extension retains the existing classical theta identity; it adds no zero-location, positive determinant, all-conductor, Lean or GRH claim. The following end-to-end adaptive readout supplies its own node-count, alias, degree, Fourier and optimization choices.

12 An end-to-end adaptive analytic readout

The new driver changes the sample truncations, interval precision, circle size and degree; it does not reuse the frozen labels to assert a smaller error. Its scope is still the three stated primitive quadratic characters and the fixed compact $|s - 1/2| \le 7/10$.

Theorem 4 (Requested-error construction in the stated finite family)**.** *Given a positive rational ε and $q \in \{3, 4, 5\}$, the specified exact-arithmetic algorithm terminates in the mathematical unbounded-resource model and returns a normalized rational polynomial P with*

$$\sup_{|s-1/2|\le 7/10} |P(s) - \Lambda_q(s)| \le \varepsilon.$$

Its sample enclosures, rational optimizer updates and all error budgets are explicit. This is an approximation theorem, with the stated classical theta identity as an input; it supplies no zero-location or positive determinant conclusion and is not a Lean theorem or a practical runtime bound.

Proof. Put $u = 7/12$, $v = 7/24$. Choose $d \ge 0$ so $10v^{d+1}/(1 - v) \le \varepsilon/8$, and choose $m > d$ so

$$10\frac{2^{-m}}{1 - 2^{-m}}\frac{1}{1 - v} \le \varepsilon/8.$$

The driver rounds m upward to a multiple of eight, which only decreases this bound. Set the requested sample error $\eta = \varepsilon(1 - u)/8$. The adaptive sampler gives all m true circle nodes and labels with error at most η. Form their outward Fourier rectangles through degree d, and refine the nodes and finite evaluation arithmetic if their weighted midpoint radii exceed $\varepsilon/8$. This refinement terminates: for these fixed finite d, m, node uncertainty and every finite-expression arithmetic width

tend to zero with precision. Merely lifting fixed boxes would not suffice, so the driver regenerates the nodes and samples on this retry.

Choose rational midpoint targets t_n and execute the previously specified quadratic-objective updates until $\sum_{n=0}^{d} |b_n^k - t_n|_1 u^n \le \varepsilon/4$. Their geometric contraction proves finite termination. The oracle, aliasing, Taylor and Fourier-midpoint terms are each at most $\varepsilon/8$, and the optimization term is at most $\varepsilon/4$. Hence the raw five-term sum is at most $3\varepsilon/4$. Choose a serialization precision B with $5\,2^{-B} \le \varepsilon/8$, round each term upward, then add the five reported rationals exactly. The printed bound is at most $7\varepsilon/8 \le \varepsilon$. The polynomial uses $((s-1/2)/(6/5))^n$, and the preceding Fourier/Cauchy proof gives its asserted compact-uniform bound. □

The trace is losslessly encoded by the fixed target vector and a rational scale extracted from each *executed* coefficient state, together with its objective, weighted residual and full-state hash. Every coefficient can be recovered exactly; a separate replay expands the states and checks every update. The producer does not generate its trace from a closed form. The saved objective remains the rational Fourier-midpoint quadratic, with its perturbation from ideal-root least squares explicitly budgeted.

13 Termination, compression and the boundary of the guarantee

The degree and circle-size choices use only rational comparisons. Since $0 < v < 1$, the Taylor test succeeds at finite d; since $2^{-m} \to 0$, the alias test succeeds at finite $m > d$. The common growth bound10 is safe for all three families; the certificate also recomputes the previously proved exact conductor-specific majorants below $4, 10, 6$.

At fixed d, m, η, the analytic truncation parameters are fixed while precision is doubled. The computed sample centers stay bounded: their underlying finite expression is fixed and the true completed-function values are bounded on the sample circle. Its arithmetic widths tend to zero, as do the regenerated true-root enclosures. Fourier computation is a finite composition of additions, products and division by the positive integer m. Its midpoint-radius sum therefore tends to zero. This proves the driver's additional precision retry, without pretending that changing a box's nominal precision shrinks its uncertainty.

The accepted sampler error may remain above its actual arithmetic width because its analytic tails and output bounds are fixed for this η. That causes no problem: the oracle allocation uses the total certified sample radius, whereas the Fourier-midpoint allocation uses only finite coefficient-rectangle width. No analytic sample error is silently moved into the latter term or removed by numerical convergence.

For the stored training trace, the exact loop gives $b^{k+1} = 3b^k/4 + t/4$ from $b^0 = 0$. Inductively every state is a rational scalar multiple of the same target vector. The producer first executes the actual vector updates, then extracts this scalar from a nonzero target coordinate, checks all coordinates, and records a canonical full-vector SHA-256 hash. If the entire target vector is zero, its zero state needs only scale0. The compressed record is thus an exact representation of the executed states, not a floating-point sketch. Hashes are integrity aids; exact vector reconstruction and the independently checked recurrence establish the mathematical claim.

Coefficient export uses exact rational numerators and denominators. A chunked decimal integer encoder/parser handles endpoints beyond Python's usual digit-conversion guard without changing global interpreter settings. No fixed 180-bit serialization floor remains. The decimal graphics are separate approximate projections, not polynomial coefficients to which the exact certificate may be applied without a conversion-error bound.

All producing source bytes are captured, copied into fresh output paths and checked unchanged. Historical fixed producers are pinned and untouched. For the new modules the run also checks the import-time source snapshots, so a stale imported module cannot be attributed to later source bytes. After output creation, a failed evaluation, final provenance check or serialization emits uncertified failure metadata. Invalid requests and stale import-time sources are refused before creating output artifacts. Old outputs are never overwritten. Extreme requests can be computationally impractical: the termination proof gives no machine-time or memory guarantee, and finite test runs establish only their reported cases.

Nothing requires the output polynomials to be real-rooted, to equal an arithmetic Fredholm determinant, or to have a self-adjoint realization. Consequently arbitrary compact approximation of these known functions does not imply GRH. Uniformity in changing conductor, general Hecke/Dedekind/elliptic/Artin scopes, the independent determinant comparison, and actual analytic/category Lean objects remain separate obligations.

14 An arithmetic-family moment with every original row retained

Let $K = \mathbb{Q}(\sqrt{-3})$, $\mathcal{O} = \mathbb{Z}[\omega]$, and write N for the ideal or element norm. Fix a finite-order Hecke character ν and a finite set S of prime ideals containing those above 6 and those dividing its conductor. For $0 \neq u \in \mathcal{O}$, let κ_u be the character of $K(u^{1/6})/K$ defined using arithmetic Frobenius, and put $\psi_u = \mathrm{prim}(\nu\kappa_u)$. Here prim means the underlying primitive Hecke character: it is formed before introducing any divisor variable or evaluating conductor-dependent zero extensions. The rows below are all actual Eisenstein elements, including units and prime powers supported in S.

The external inputs are abelian Kummer reciprocity and character uniqueness, the primitive Hecke functional equation and its usual vertical growth, Stirling's formula, scalar Dedekind-zeta convexity, and Blomer–Goldmakher–Louvel's large sieve for their specified sixth-order family. We use the pinned arXiv:1112.1650v1, Theorem 1.3, Lemma 2.1 and Appendix II; its Section 5 supplies the all-ideal decomposition whose fixed-twist proof is redone here. No statement about all primitive characters of comparable conductor, Artin holomorphy, Lindelöf bounds or zero locations is assumed.

Theorem 5 (The original-family numerator moment)**.** *For every $\varepsilon > 0$, uniformly for $H \geq 1$ and $t \in \mathbb{R}$,*

$$\sum_{0<Nu\leq H} |L(1/2+it,\psi_u)|^2 \ll_{\nu,S,\varepsilon} H^{1+\varepsilon}(1+|t|)^{1+\varepsilon}. \tag{1}$$

For every fixed $0 < \delta < 1/2$, the same estimate holds with $1/2+it$ replaced by $1/2+\delta+it$, with the constant also depending on δ. If the central bound has constant C_0 at an exponent $0 < \varepsilon_0 \leq 1/4$, an admissible off-central constant is

$$200C_0\left(\frac{3/2+\delta}{1/2-\delta}\right)^2. \tag{2}$$

Proof. For the canonical BGL character $\vartheta_{\mathfrak{a}}$, Appendix II gives a radicand $\alpha_{\mathfrak{a}}$ with $(\alpha_{\mathfrak{a}})_S = \mathfrak{a}\mathfrak{g}^{-6}$. Since $\mathcal{O}_S$ is principal, comparison with any literal u whose outside-S ideal is $\mathfrak{a}$ gives $\kappa_u = \eta\vartheta_{\mathfrak{a}}$ for one of finitely many fixed S-unit Kummer twists η. This is an equality of primitive characters, established outside a finite set and extended by uniqueness. For each outside-S ideal $\mathfrak{a}$ and S-supported integral ideal $\mathfrak{b}$, at most six elements generate $\mathfrak{a}\mathfrak{b}$. The resulting S-supported multiplicity sum converges at exponent $1+\varepsilon$.

For each fixed $\xi = \nu\eta$, split the canonical parameter as $\mathfrak{a} = \mathfrak{a}_0\mathfrak{a}_1\mathfrak{a}_2^6$, where $\mathfrak{a}_0$ is squarefree, $\mathfrak{a}_1$ is squarefull and sixth-power-free, and $(\mathfrak{a}_0, \mathfrak{a}_1) = 1$. The primitive conductor is at most a fixed multiple of $N(\mathfrak{a}_0\mathfrak{a}_1)$. The common-gamma approximate functional equation, with $X = H/(N\mathfrak{a}_2)^6$

and $T = 1 + |t|$, has common maximal length $B = X^{1/2}T(HT)^{\eta}$ for any sufficiently small fixed $\eta > 0$. Mellin inversion removes its row-dependent conductor weight. Split a column as $\mathfrak{b}_S\mathfrak{b}_1\mathfrak{b}_2$, with $\mathfrak{b}_S$ supported in S, $\mathfrak{b}_1$ squarefree and $\mathfrak{b}_2$ squarefull, the last two coprime and prime to S. The original primitive values at $\mathfrak{b}_S$ remain bounded row multipliers. The S-supported and squarefull column weights are summable at $1/2 + \eta$. Multiplying squarefree column coefficients by the fixed ξ preserves their squared norm, whatever its finite order. BGL's large sieve therefore bounds the energy for fixed $\mathfrak{a}_1, \mathfrak{a}_2$ by

$$(HT)^{O(\eta)}\{M + B + (MB)^{2/3}\}, \qquad M = X/N\mathfrak{a}_1.$$

Squarefull counting and summation give $X + B\sqrt{X} + (XB)^{2/3} \ll (HT)^{O(\eta)}XT$. The $\mathfrak{a}_2$ sum converges at exponent 6. Principal rows are separated before the pole-free functional equation and bounded by scalar zeta convexity. This proves (1).

For the off-central statement divide each $L(s, \psi)$ by $(s + 1)^{5/4}$ and, on principal rows, also multiply by $(s - 1)/(s + 1)$. These are bounded analytic functions on $\operatorname{Re} s > 1/2$, continuous on the boundary. Poisson representation and Jensen's inequality transfer the central squared moment. The convolution of $P_\delta(y) = \delta/[\pi(\delta^2 + y^2)]$ with $(1 + |t|)^{-3/2+\varepsilon_0}$ is at most eleven times that weight when $0 < \varepsilon_0 \le 1/4$. Restoring the denominator and the principal pole-removal factor gives (2). The extended edition gives the detailed derivation with all primitive local values. □

Separate fixed-unitary extension. The moment theorem also holds for any one fixed unitary Hecke character ν. For precision choose $\nu(\mathfrak{a}) = \nu_0(\mathfrak{a})(N\mathfrak{a})^{-i\tau}$, with angular parameter $m \in \mathbb{Z}$ and completed gamma factor

$$\gamma_\nu(s) = 2(2\pi)^{-(s+|m|/2+i\tau)}\Gamma(s + |m|/2 + i\tau).$$

All finite Kummer twists have this same infinity type. Use $T_\nu = 1 + |t + \tau| + |m|$, comparable to $1 + |t|$ with ν-dependent constants. Pure norm rows have $L(s, \psi) = \zeta_K(s + i\tau)$ and are separated before the pole-free functional equation; their pole at $1 - i\tau$ is removed by $(s - 1 + i\tau)/(s + 1 + i\tau)$. No uniformity in m, τ or the conductor of ν is asserted. This extension of the moment is separate from the following finite-order type-I residue formula.

14.1 The exact masked remainder and the remaining loss

Retain the original zero-extended symbol $\chi_d(u) = (u/d)_6$ on good squarefree ideals. Let μ be the ideal Möbius function and set

$$\beta_U(d) = \sum_{\substack{ab=d \\ Na, Nb \le U}} \mu(a)\mu(b), \quad T(u, d) = S \cup \operatorname{supp}(u) \cup \operatorname{supp}(d).$$

Let $\psi_{u,2} = \operatorname{prim}(\psi_u^2)$, formed independently of squaring any zero sentinel. With $\mathfrak{f}_u$ and $\mathfrak{f}_{u,2}$ the primitive conductors, the full-mask series is

$$F_{u,d}(s) = \frac{L(s, \psi_u)}{L(2s, \psi_{u,2})} C_{T(u,d)}(s), \tag{3}$$

$$C_T(s) = \frac{\prod_{p \in T,\ p \nmid \mathfrak{f}_u}(1 - \psi_u(p)(Np)^{-s})}{\prod_{p \in T,\ p \nmid \mathfrak{f}_{u,2}}(1 - \psi_{u,2}(p)(Np)^{-2s})}.$$

Initially it equals $\sum_{c \text{ squarefree},\ (c,T)=1} \psi_u(c)(Nc)^{-s}$ for $\operatorname{Re} s > 1$. Every prime of T remains deleted, including those forgotten by either primitive conductor.

For finite-order ν, $D \geq 1$, $U \geq 1$, $U^2 \leq D$, and $W \in C_c^\infty((0,\infty))$ supported in $[1,2]$, put

$$\widehat{W}(s) = \int_0^\infty W(x) x^{s-1}\, dx, \qquad \mathcal{M}_4(W) = \sum_{j=0}^{4} \sup_x |(x\partial_x)^j W(x)|.$$

On $\sigma = 1/2 + \delta$, the exact residue-subtracted type-I remainder is

$$E_u = \frac{D^\delta}{2\pi} \int_{\mathbb{R}} \widehat{W}(\sigma + it) D^{it} \sum_{\substack{d \text{ squarefree} \\ (d,S)=1}} \frac{\beta_U(d)\nu(d)\chi_d(u)}{(Nd)^{\sigma+it}} F_{u,d}(\sigma + it)\, dt. \tag{4}$$

This definition retains the actual β_U, all phases, d-masks, conductors and the independently primitive square.

Theorem 6 (An improved masked residual estimate)**.** *Under the preceding finite-order hypotheses, for every $\varepsilon > 0$,*

$$\sum_{0<Nu\leq H} |E_u|^2 \ll_{\nu,S,\delta,\varepsilon} \mathcal{M}_4(W)^2 D^{2\delta} U^{2-4\delta} H (HD)^\varepsilon. \tag{5}$$

Proof. The same deleted set in both factors of (3) gives $|F_{u,d}(\sigma + it)| \ll (Nu\, Nd)^\eta |L(\sigma + it, \psi_u)|$: the reciprocal incomplete Euler product at $2\sigma > 1$ is bounded by $\zeta_K(2\sigma)$, and its remaining finite numerator is bounded by a small power of its radical. Apply Minkowski to (4) in the exact row space and use Theorem 5. The actual coefficient sum is at most

$$\sum_d \frac{|\beta_U(d)|}{(Nd)^{\sigma-\eta}} \leq \left(\sum_{Na\leq U} (Na)^{-\sigma+\eta} \right)^2 \ll U^{1-2\delta+2\eta}.$$

Four Mellin integrations by parts bound the remaining height integral by a constant times $\mathcal{M}_4(W)$. Square the result and absorb the small losses using $U^2 \leq D$. □

At $H = D^{3/4}$ and $U = D^a$ with fixed $0 < a \leq 1/2$, the upper bound's excess over H is $2\delta + (2 - 4\delta)a$, up to small losses; as $\delta \downarrow 0$ it still leaves $2a$. This is a limitation of the upper bound, not an energy lower bound. The extracted principal residue, the original short term at smaller scales and the signed type-II minus type-I covariance are not controlled by (5). The next arithmetic task is to estimate the complete d-weighted moment before its absolute coefficient sum, or establish sufficient signed covariance. No positive arithmetic determinant, zero-location theorem or GRH proof follows. The analytic argument uses the external classical inputs above; finite diagnostics and selected existing Lean local identities do not formalize the Hecke functions, approximate functional equation, Poisson transfer or these moment bounds.

15 Family identification, the approximate functional equation and masks

We prove the analytic statements of Section 14 from the explicit classical inputs stated there. Write $I(S)$ for the nonzero integral ideals prime to S, and $\mathcal{O}_S$ for the ring of S-integers. Constants may depend on the fixed field, S, ν and any declared positive small exponent. No such constant is asserted uniform in the conductor or infinity type of a varying ν.

15.1 A finite comparison with the literal Kummer rows

Use Appendix II of BGL, with $n = 6$. Its fixed local modulus and representatives may be chosen once, since S contains the primes above 6 and $\mathcal{O}_S$ has class number one. For each $\mathfrak{a} \in I(S)$ the construction writes

$$\mathfrak{a} = (x)E\mathfrak{g}^6, \qquad \alpha_{\mathfrak{a}} = xm_E, \qquad (m_E)_S = E,$$

so $(\alpha_{\mathfrak{a}})_S = \mathfrak{a}\mathfrak{g}^{-6}$. Choose $y \in K^\times$ generating $\mathfrak{g}$ as an $\mathcal{O}_S$-ideal. If $(u)_S = \mathfrak{a}$, then

$$e = \alpha_{\mathfrak{a}} y^6/u \in \mathcal{O}_S^\times, \qquad [\alpha_{\mathfrak{a}}] = [eu] \quad \text{in } K^\times/K^{\times 6}.$$

There is a precision here: Appendix II allows choosing $\mathfrak{g}$ relative to a tested denominator. Fix one admissible global decomposition first. Its radicand gives the canonical character $\vartheta_{\mathfrak{a}}$ on all primes outside the finite set $S \cup \mathrm{supp}(\mathfrak{a}\mathfrak{g})$ where that choice applies. Uniqueness of Hecke characters, equivalently Chebotarev in the finite abelian quotient, then identifies the underlying primitive characters everywhere. With matching arithmetic-Frobenius conventions,

$$\kappa_u = \kappa_{e^{-1}}\vartheta_{\mathfrak{a}}. \tag{6}$$

If the residue-symbol convention is conjugated, conjugate the canonical family and its coefficient sequence; the same large sieve follows.

The group $\mathcal{O}_S^\times/(\mathcal{O}_S^\times)^6$ is finite. Thus (6) uses one of a finite set of fixed Kummer twists η, all unramified outside S and of bounded S-conductor. This is a comparison of primitive characters, not a pointwise multiplication rule for independently zero-extended arrays at bad primes. In particular, a conductor drop must not be replaced by a product of artificial zeros.

Since $\mathcal{O}$ is principal with six units, for fixed $\mathfrak{a} \in I(S)$ and an S-supported integral ideal $\mathfrak{b}$ there are at most six u satisfying $(u) = \mathfrak{a}\mathfrak{b}$. For any nonnegative row quantity, the literal moment is consequently at most

$$6 \sum_{\substack{\mathfrak{b}\ S\text{-supported} \\ N\mathfrak{b} \le H}} \sum_{\eta} \sum_{\substack{\mathfrak{a} \in I(S) \\ N\mathfrak{a} \le H/N\mathfrak{b}}} \left|L(s, \mathrm{prim}(\nu\eta\vartheta_{\mathfrak{a}}))\right|^2. \tag{7}$$

An ideal bound $X^{1+\varepsilon}T^{1+\varepsilon}$ therefore transfers to all literal rows, since

$$\sum_{\mathfrak{b}\ S\text{-supported}} (N\mathfrak{b})^{-1-\varepsilon} = \prod_{p \in S} (1 - (Np)^{-1-\varepsilon})^{-1} < \infty.$$

No quotient of the original unit rows or deletion of S-prime powers has occurred.

15.2 Fixed twists, exact tame conductors and the squarefree sieve

Fix $\xi = \nu\eta$ and write $\rho_{\mathfrak{a}} = \mathrm{prim}(\xi\vartheta_{\mathfrak{a}})$. Its fixed conductor is supported in S before the Kummer twist. On columns prime to S, $|\xi(\mathfrak{b})| = 1$. BGL Theorem 1.3 thus gives, for any fixed $\mathfrak{a}_1$,

$$\sum_{\substack{\mathfrak{a}_0 \in I(S)\ \text{squarefree} \\ N\mathfrak{a}_0 \le M}} \left| \sum_{\substack{\mathfrak{b}_1 \in I(S)\ \text{squarefree} \\ N\mathfrak{b}_1 \le B}} \lambda_{\mathfrak{b}_1} \xi(\mathfrak{b}_1)\vartheta_{\mathfrak{a}_1}(\mathfrak{b}_1)\vartheta_{\mathfrak{a}_0}(\mathfrak{b}_1) \right|^2$$
$$\ll (MB)^{\varepsilon}\{M + B + (MB)^{2/3}\} \sum_{\mathfrak{b}_1} |\lambda_{\mathfrak{b}_1}|^2. \tag{8}$$

The first fixed multiplier preserves the coefficient norm; the second has modulus at most one. Restricting the positive outer row sum is harmless. A coprimality condition with another fixed ideal

belongs to λ. In particular, ν need not have order dividing six and is not asserted to belong to BGL's finite ray-class character set.

There is a unique valuation decomposition

$$\mathfrak{a} = \mathfrak{a}_0\mathfrak{a}_1\mathfrak{a}_2^6, \quad \mathfrak{a}_0 \text{ squarefree}, \quad \mathfrak{a}_1 \text{ squarefull and sixth-power-free}, \quad (\mathfrak{a}_0, \mathfrak{a}_1) = 1.$$

Indeed, for $v_p(\mathfrak{a}) = 6q + r$, $0 \le r \le 5$, put the residual prime in $\mathfrak{a}_0$ if $r = 1$, and its exponent r in $\mathfrak{a}_1$ if $2 \le r \le 5$. BGL Lemma 2.1 gives $\vartheta_{\mathfrak{a}\mathfrak{b}^6} = \vartheta_{\mathfrak{a}}$ and $\vartheta_{\mathfrak{a}_0\mathfrak{a}_1} = \vartheta_{\mathfrak{a}_0}\vartheta_{\mathfrak{a}_1}$ on ideals prime to S in this sixth-power-free case. Those are the identities needed here; no arbitrary multiplicativity assertion is substituted.

At a prime outside S, tame Kummer inertia at residual exponent r has order $6/\gcd(6, r)$. It is nontrivial for $r = 1, \dots, 5$ and has conductor exponent one. The fixed ξ is unramified there, so cannot cancel that inertia. At primes in S the local sixth-power classes are finite, hence only finitely many primitive conductor types occur after multiplication by the fixed ξ. Consequently

$$Q_{\rho_{\mathfrak{a}}} = |D_K| N\mathfrak{f}_{\rho_{\mathfrak{a}}} \le C_{\xi,S} N(\mathfrak{a}_0\mathfrak{a}_1), \tag{9}$$

independently of $\mathfrak{a}_2$. The equality of the conductor's outside-S part with $\mathrm{rad}(\mathfrak{a}_0\mathfrak{a}_1)$ does not identify that conductor with the original larger deleted set.

15.3 Removing the row-dependent functional-equation weights

First take finite-order ν. All characters then have trivial infinity type at the complex place. The primitive completion may be normalized as

$$\Lambda(s, \rho) = Q_\rho^{s/2}\gamma_K(s)L(s, \rho), \qquad \gamma_K(s) = 2(2\pi)^{-s}\Gamma(s), \qquad \Lambda(s, \rho) = \epsilon_\rho\Lambda(1 - s, \bar{\rho}), \quad |\epsilon_\rho| = 1.$$

For nonprincipal rows the completed function is entire. Apply this functional equation to the Mellin integral with even auxiliary factor $G(z) = e^{z^2}$ and denominator z, shifting across $z = 0$. At $s = 1/2 + it$ this gives two weighted Dirichlet series, truncated below to polynomials; the dual prefactor has modulus one. Principal rows are excluded from this pole-free step and treated separately below.

For fixed $\mathfrak{a}_2$, let $X = H/(N\mathfrak{a}_2)^6 \ge 1$ and $T = 1 + |t|$. An explicit normalized kernel is

$$V_t(x) = \frac{1}{2\pi i}\int_{(\eta)} e^{z^2}\frac{\Gamma(s + z)}{\Gamma(s)}T^{-z}x^{-z}\,\frac{dz}{z}, \qquad x = \frac{2\pi N\mathfrak{b}}{\sqrt{Q_\rho}T}, \tag{10}$$

with $\eta > 0$ small. Its factors combine to $(\sqrt{Q_\rho}/(2\pi N\mathfrak{b}))^z$ in the actual completed-L kernel; there is no extra factor of T. The denominator z supplies the residue at zero. On $z = \eta + iy$, Stirling and $|t + y| \le T(1 + |y|)$ give

$$\left|\frac{\Gamma(1/2 + \eta + i(t + y))}{\Gamma(1/2 + it)T^{\eta+iy}}\right| \ll_\eta (1 + |y|)^{C_\eta}e^{\pi|y|/2}.$$

The factor e^{-y^2} makes this an integrable majorant uniformly in t. The analogous bound holds on every fixed positive shifted line and for the dual gamma kernel. Moving far enough to the right gives arbitrary power decay beyond a common total-ideal cutoff

$$B' = X^{1/2}T(HT)^\eta, \tag{11}$$

up to fixed ξ, S, K constants. For any requested $A > 0$, a sufficiently large positive kernel shift, depending also on η, makes the truncation error $O((HT)^{-A})$ uniformly in all rows in this block.

Ideal counting $O_K(Y)$ bounds the absolute tails. Such errors are negligible after the finite row sums by taking A larger if needed.

Insert (10) on $\operatorname{Re} z = \eta$. The factor $(\sqrt{Q_\rho}T)^z$ is independent of the column and is a row multiplier of modulus at most $(CXT^2)^{\eta/2}$. Its imaginary part is a phase. The remaining column coefficient is $(N\mathfrak{b})^{-1/2-\eta-it-iy}$, apart from fixed constants. Minkowski or Cauchy with the integrable gamma majorant therefore removes the row-dependent weight at cost $(HT)^{O(\eta)}$ in the squared norm. No independence of the root numbers is assumed or needed.

For each column use the unique decomposition

$$\mathfrak{b} = \mathfrak{b}_S\mathfrak{b}_1\mathfrak{b}_2, \quad \mathfrak{b}_S \ S\text{-supported}, \quad \mathfrak{b}_1 \text{ squarefree}, \quad \mathfrak{b}_2 \text{ squarefull}, \quad (\mathfrak{b}_1, \mathfrak{b}_2) = 1,$$

where $\mathfrak{b}_1, \mathfrak{b}_2 \in I(S)$; a prime of outside-$S$ valuation one goes in $\mathfrak{b}_1$, and a prime of valuation at least two goes with its full exponent in $\mathfrak{b}_2$. For fixed $\mathfrak{b}_S, \mathfrak{b}_2$ retain $\rho_{\mathfrak{a}}(\mathfrak{b}_S)$ as the original primitive row value. On the good factors the exact canonical identity applies. The remaining factors involving $\mathfrak{b}_2$, including $\vartheta_{\mathfrak{a}_0}(\mathfrak{b}_2)$, are bounded row multipliers. Their moduli are at most one; only then are they discarded in an inequality. In particular the value at $\mathfrak{b}_S$ is not replaced by the product of zero-extended primitive arrays.

For the $\mathfrak{b}_1$ polynomial the coefficient sequence is independent of $\mathfrak{a}_0$; it includes the indicator $(\mathfrak{b}_1, \mathfrak{b}_2) = 1$ and the common cutoff $N\mathfrak{b}_1 \le B'/N(\mathfrak{b}_S\mathfrak{b}_2)$. Padding it by zeros up to B' is legitimate. Its squared norm is at most $\zeta_K(1+2\eta)$. Minkowski over $\mathfrak{b}_S, \mathfrak{b}_2$ is legitimate because

$$\sum_{\mathfrak{b}_S, \mathfrak{b}_2} N(\mathfrak{b}_S\mathfrak{b}_2)^{-1/2-\eta} < \infty.$$

The finite Euler product handles $\mathfrak{b}_S$; the squarefull series converges above $1/2$. Apply (8) to each such polynomial and to the conjugate dual family. For fixed $\mathfrak{a}_1, \mathfrak{a}_2$ the resulting energy is

$$(HT)^{O(\eta)}\{M + B' + (MB')^{2/3}\}, \qquad M = X/N\mathfrak{a}_1. \tag{12}$$

The additional condition $(\mathfrak{a}_0, \mathfrak{a}_1) = 1$ simply restricts the positive row sum. The large-sieve small exponent can be chosen sufficiently small along with η and absorbed in the indicated small loss.

15.4 All-ideal summation and principal rows

Every squarefull ideal $\mathfrak{c}$ has a unique expression $\mathfrak{d}^2\mathfrak{e}^3$ with $\mathfrak{e}$ squarefree. This follows primewise by using exponent zero or one in $\mathfrak{e}$ according to the parity of an exponent at least two. Therefore

$$\#\{\mathfrak{c} \text{ squarefull} : N\mathfrak{c} \le X\} \ll_K X^{1/2}, \qquad \sum_{\mathfrak{c} \text{ squarefull}} (N\mathfrak{c})^{-s} < \infty \quad (s > 1/2).$$

For the count, sum $O((X/(N\mathfrak{e})^3)^{1/2})$ and use $\zeta_K(3/2) < \infty$. Summing (12) over $\mathfrak{a}_1$ gives

$$(HT)^{O(\eta)}\{X + B'\sqrt{X} + (XB')^{2/3}\} \ll (HT)^{O(\eta)}XT,$$

since the squarefull series converges at 1 and $2/3$, $B' = \sqrt{X}\,T(HT)^\eta$, and $T^{2/3} \le T$ for $T \ge 1$. The $\mathfrak{a}_2$ sum is bounded by $H\sum_{\mathfrak{a}_2}(N\mathfrak{a}_2)^{-6} = H\zeta_K(6)$. Choosing all internal losses small for the requested final ε proves the fixed-twist all-ideal central moment.

There are $O_K(H)$ ideal rows in total. On any row where the primitive character is principal its L-function is ζ_K, and ordinary scalar convexity gives $|\zeta_K(1/2+it)|^2 \ll_\varepsilon T^{1+\varepsilon}$. This treats those rows without a pole-free approximate functional equation. Finally (7) proves (1) for every actual nonzero Eisenstein element.

15.5 Poisson transfer with an explicit constant

For a nonprincipal row define

$$f_\psi(s) = \frac{L(s,\psi)}{(s+1)^{5/4}},$$

using the holomorphic logarithm on $\mathrm{Re}(s+1) > 0$. On a principal row additionally multiply by $(s-1)/(s+1)$. Analytic continuation and ordinary degree-two convexity, used separately at each fixed conductor, show that f_ψ is bounded analytic on the right half-plane $\mathrm{Re}\, s > 1/2$, with continuous boundary and decay at infinity. The principal pole is removed before applying Poisson representation to its real and imaginary parts. Jensen gives

$$|f_\psi(1/2+\delta+it)|^2 \le \int_{\mathbb{R}} P_\delta(t-v)|f_\psi(1/2+iv)|^2\,dv.$$

The exact row family is finite for fixed H, so its sum interchanges with the integral without a limiting-family issue. The boundary pole-removal factor has modulus at most one.

Take $0 < \varepsilon_0 \le 1/4$ and $\alpha = 3/2 - \varepsilon_0 \in [5/4, 3/2)$. Since $|3/2+iv| \ge (1+|v|)/\sqrt{2}$, the central squared moment of the f_ψ is bounded by $3C_0H^{1+\varepsilon_0}w_\alpha(v)$, where $w_\alpha(v) = (1+|v|)^{-\alpha}$. For $|t| \ge 2$, split $|v-t| \le |t|/2$ from its complement. The near contribution to $P_\delta * w_\alpha$ is at most $2^\alpha w_\alpha(t) < 3w_\alpha(t)$. In the far region use $P_\delta(t-v) \le 4\delta/(\pi t^2)$ and $\int w_\alpha = 2/(\alpha-1)$, giving at most

$$\frac{2\delta 3^\alpha}{\pi(\alpha-1)} w_\alpha(t) \le 8w_\alpha(t).$$

Here $(1+|t|)^\alpha/t^2 \le 3^\alpha/4$ for $|t| \ge 2$. For $|t| \le 2$, the convolution is at most one and hence at most $3^\alpha w_\alpha(t) < 6w_\alpha(t)$. Thus $P_\delta * w_\alpha \le 11 w_\alpha$ uniformly for $0 < \delta < 1/2$. Restoring the denominator costs less than $6(1+|t|)^{5/2}$, and the inverse principal factor is bounded by $(3/2+\delta)/(1/2-\delta)$. The product $11 \cdot 3 \cdot 6 = 198 < 200$ proves (2). The constant is uniform as $\delta \downarrow 0$, but not as the contour approaches the principal pole line $\mathrm{Re}\, s = 1$. No zero-free hypothesis appears.

15.6 The separately scoped unitary infinity type

Let $\nu(\mathfrak{a}) = \nu_0(\mathfrak{a})(N\mathfrak{a})^{-i\tau}$ be fixed and unitary, with angular parameter m and gamma factor γ_ν stated in Section 14. The external primitive functional equation is explicitly

$$Q_\rho^{s/2}\gamma_\nu(s)L(s,\rho) = \epsilon_\rho Q_\rho^{(1-s)/2}\gamma_{\bar\nu}(1-s)L(1-s,\bar\rho), \qquad |\epsilon_\rho| = 1.$$

A constant norm-twist phase in another completion convention is absorbed in ϵ_ρ. All finite Kummer twists share this infinity type. Replace T in the kernel by $T_\nu = 1 + |t+\tau| + |m|$. The numerator gamma ratio now has real base $1/2 + |m|/2 > 0$ and vertical parameter $t+\tau$; its conjugate dual has parameter $-(t+\tau)$. Stirling gives the same integrable Gaussian-majorized kernel on each fixed shifted line. Constants can depend on m, τ; T_ν is comparable to T for this one fixed ν. The finite-prime conductor analysis, modulus-one coefficient twist, all-ideal decomposition and all original primitive row values remain unchanged.

A pure norm row has $L(s,\rho) = \zeta_K(s+i\tau)$; exclude it from the pole-free approximate functional equation and use scalar convexity at $t+\tau$. In the Poisson step cancel its actual pole by

$$h_\tau(s) = \frac{s-1+i\tau}{s+1+i\tau}.$$

This factor is analytic in the relevant right half-plane. On its boundary $|h_\tau| \le 1$, while inside $|h_\tau^{-1}| \le (3/2+\delta)/(1/2-\delta)$. Hence the same transfer factor 200 applies to the now ν-dependent

central constant. If the opposite norm convention is chosen, replace τ by $-\tau$ in the gamma factor and pole alike. This proves the separate fixed-unitary moment assertion; it does not claim uniformity in a family of infinity types or replace the finite-order principal-residue expression below.

15.7 Keeping the full type-I mask during transfer

For $\sigma = 1/2 + \delta$, define L^T by removing every prime of $T(u,d)$, not just conductor primes. The identity (3) is equivalently

$$F_{u,d}(s) = \frac{L(s,\psi_u)}{L^{T(u,d)}(2s,\psi_{u,2})} \prod_{p \in T(u,d),\ p \nmid \mathfrak{f}_u} (1 - \psi_u(p)(Np)^{-s}).$$

Absolute convergence at $2\sigma > 1$ bounds the reciprocal incomplete product by $\zeta_K(2\sigma)$, without requiring any zero-free theorem in the critical strip. For every fixed $\eta > 0$,

$$\prod_{p \in T} (1 + (Np)^{-\sigma}) \ll_{\sigma,\eta} (N \operatorname{rad} T)^{\eta}, \qquad N \operatorname{rad} T \le C_S N u\, N d.$$

Indeed $1 + q^{-\sigma} \le q^{\eta}$ beyond a fixed threshold, and only finitely many prime ideals lie below it. Therefore

$$|F_{u,d}(\sigma + it)| \ll_{\nu,S,\delta,\eta} (N u\, N d)^{\eta} |L(\sigma + it, \psi_u)|. \tag{13}$$

This is a bounded inequality after the exact mask identity, not an identity that removes original zeros. For example, with principal ν and $u = \pi^3$ or π^6 at $N\pi = 7$, the primitive local ratio, restoration and full masked factor at $s = 1$ are respectively

$v_\pi(u)$	primitive ratio	restoration	masked factor
3	48/49	49/48	1
6	8/7	7/8	1

The squared primitive character in the first row is unramified at π although the original primitive character is ramified there. Squaring a zero sentinel would give the wrong denominator.

For finite-order ν, the normalized type-I piece is the retained d sum of the smoothed squarefree series at $Y = D/Nd$, divided by $\sqrt{D}$. Its possible pole is only at $s = 1$, on principal rows. Contour shifting with the full mask gives $I_u = P_u + E_u$, with E_u exactly (4) and

$$P_u = \mathbf{1}_{\psi_u = 1} \sqrt{D}\, \widehat{W}(1) \sum_{\substack{d \text{ squarefree} \\ (d,S)=1}} \frac{\beta_U(d)\nu(d)\chi_d(u)}{Nd} \rho_{T(u,d)},$$

$$\rho_T = \frac{\operatorname{res}_{s=1} \zeta_K(s)}{\zeta_K(2)} \prod_{p \in T} (1 + (Np)^{-1})^{-1}.$$

This does not impose positivity on P_u. Since β_U is supported on $Nd \le U^2 \le D$, every inner scale Y is at least one. Four integrations by parts give $|\widehat{W}(\sigma + it)| \ll_\sigma \mathcal{M}_4(W)(1 + |t|)^{-4}$, which justifies the shifted integrals and their horizontal limits with the classical Hecke growth estimates.

Apply Minkowski to (4) in ℓ^2 of the actual u rows, then use (13) and the off-central moment. For sufficiently small internal losses this bounds $\|E\|_2$ by

$$C D^{\delta} H^{1/2 + O(\eta)} \sum_d \frac{|\beta_U(d)|}{(Nd)^{\sigma - \eta}} \int_{\mathbb{R}} |\widehat{W}(\sigma + it)| (1 + |t|)^{(1 + O(\eta))/2}\, dt.$$

All original $\nu(d)\chi_d(u)$ have modulus at most one and remain inside the exact expression before this inequality. The actual truncated factorization gives, without replacing β_U by arbitrary coefficients,

$$\sum_d \frac{|\beta_U(d)|}{(Nd)^{\sigma-\eta}} \le \left(\sum_{Na\le U} (Na)^{-\sigma+\eta}\right)^2 \ll_{\delta,\eta} U^{1-2\delta+2\eta}.$$

Choose $\eta < \delta$ if necessary and use the fixed-field ideal count. The height integral is bounded by $C\mathcal{M}_4(W)$. Squaring and absorbing the small losses in $(HD)^{\varepsilon}$, using $U^2 \le D$, proves (5). Thus the formerly conditional numerator moment input in this residual transfer is discharged, while its positive-power cutoff loss is retained explicitly.

15.8 What remains open and what has been verified

The new remainder envelope improves the old scalar-convexity H power $3/2 - \delta + \varepsilon$ to $1 + \varepsilon$. At fixed U one may choose δ for a requested small-loss exponent; for $U = D^a$ with fixed $a > 0$, the remaining excess tends to $2a$ as $\delta \downarrow 0$. Neither statement controls P_u, the other signed piece, nor their covariance. In the original Vaughan identity the short $2a_U$ term must be restored at smaller scales where its support can meet the annulus. Uniform $\mathcal{M}_4$ bounds do not justify interchanging a profile supremum with the row moment or assuming $U^2 \le D'$ after a scale D' is reduced.

The independent prose audit is recorded in `planning/analytic-oracle/arithmetic-review/adversarial-afm-review-20261104.md`. The source-hashed finite checker tests valuation/support decompositions, squarefull coordinates, height exponents and rational Poisson envelopes. Its finite success does not prove reciprocity, analytic continuation, the large sieve, the approximate functional equation or the moments. No new Lean declarations accompany these analytic theorems. Existing Lean local Euler identities are a separately limited verification layer. The independent positive arithmetic determinant comparison, original signed low-gcd short-row estimate, all required contour uniformities and GRH remain open; Dirichlet, Hecke, Dedekind and general Artin scopes are not identified with one another.

Pinned primary sources and external inputs. The principal analytic family input is Blomer–Goldmakher–Louvel, arXiv:1112.1650v1, Theorem 1.3, Lemma 2.1, Appendix II and Section 5. Our fixed-ν proof changes column coefficients and redoes the functional-equation reduction; it does not claim the published corollary directly states every fixed twist. For comparison, Gao–Zhao, arXiv:2201.01885v2, Lemma 2.9 and Section 3.2, uses the sextic large-sieve/moment interface. Its GRH-conditional central-density theorem is not invoked. Abelian Kummer identification uses the classical reciprocity theorem, with the project's pinned Milne *Class Field Theory*, version 4.03 (6 August 2020), Theorem 0.8 and Chapter I, Section 1, a reference: Milne's notes. The primitive degree-one Hecke continuation and completed functional equation are explicit classical external inputs; the project's pinned Kudla notes on Tate's thesis, Theorem 4.3 and Corollary 4.4, supply a reference. Their sketch and incomplete displayed epsilon formulas are not used as independently certified constants. Standard Stirling, Mellin inversion, Poisson representation and scalar convexity complete the external analytic dependencies. None is replaced by a finite diagnostic or a GRH-equivalent assumption.

16 Recombining the actual weighted cutoff

Keep the finite-order Eisenstein family and original masks of Section 14. For $U \geq 1$ and a squarefree ideal m, put

$$A_U(m) = \sum_{d|m} \beta_U(d) = \sum_{\substack{ab|m \\ (a,b)=1,\ Na,Nb\leq U}} \mu(a)\mu(b).$$

Every summation variable here is an integral ideal. The last formula uses squarefreeness of m; an unconstrained square of a Möbius polynomial would also include forbidden overlapping supports.

Theorem 7 (Exact recombination and an absolute-half-plane rate)**.** *Write* $T_0 = S \cup \mathrm{supp}(u)$ *and* $J_{u,U}(s) = \sum_{d \text{ squarefree},\ (d,S)=1} \beta_U(d)\nu(d)\chi_d(u)(Nd)^{-s}F_{u,d}(s)$*. For* $\mathrm{Re}\, s > 1$*,*

$$J_{u,U}(s) = \sum_{m \text{ squarefree},\ (m,T_0)=1} A_U(m)\psi_u(m)(Nm)^{-s},$$
$$|A_U(m)| \leq 3^{\omega(m)}, \qquad A_U(m) = \mu(m) \quad (Nm \leq U).$$

For every $\kappa > 0$ *and* $0 < \epsilon < \kappa$*, uniformly for* $\mathrm{Re}\, s \geq 1 + \kappa$*, all original rows and all* $U \geq 1$*,*

$$\left| J_{u,U}(s) - \frac{1}{L^{T_0}(s,\psi_u)} \right| \leq U^{-\kappa+\epsilon}\{\zeta_K(1+\epsilon)^3 + \zeta_K(1+\epsilon)\}.$$

Here L^{T_0} *removes every prime of* T_0*, including conductor-drop primes; the bound is in the absolute-convergence half-plane only.*

Proof. On a surviving term $m = dc$, the two squarefree ideals c, d are coprime, and both avoid T_0. The original product $\nu(d)\chi_d(u)\psi_u(c)$ is then $\psi_u(m)$. Grouping by m gives A_U. Each prime of m has three possible positions in $(a, b, m/(ab))$, so the absolute sum is at most $3^{\omega(m)}$. When $Nm \leq U$ all divisors fit both cutoffs; the primewise sum is $1 - 1 - 1 = -1$, giving $\mu(m)$.

The same bound justifies absolute convergence and regrouping for $\mathrm{Re}\, s > 1$, because on squarefree ideals $3^{\omega(m)} = \tau_3(m)$ and $\sum_m \tau_3(m)(Nm)^{-v} = \zeta_K(v)^3$ for $v > 1$, where this latter sum runs over all nonzero integral ideals. Subtract the absolutely convergent Möbius series for $1/L^{T_0}$. Only $Nm > U$ remain. Rankin's inequality with exponent $\eta = \kappa - \epsilon$ bounds their absolute sum by

$$U^{-\eta} \sum_m \{\tau_3(m) + 1\}(Nm)^{-1-\epsilon},$$

which proves the stated constant independently of deleted primes, row multiplicities or character phases. □

For W supported in $[1, 2]$, the same exact regrouping gives the original normalized type-I polynomial

$$I_u = D^{-1/2} \sum_{m \text{ squarefree},\ (m,S)=1} A_U(m)\nu(m)\chi_m(u)W(Nm/D).$$

This formula retains all original zero sentinels. It permits a direct family large sieve before any absolute d sum; its quantitative moment is developed separately below. The displayed absolute-half-plane rate alone is not that critical-strip moment.

17 Euler packaging, coprimality and the analytic boundary

For $p \notin T_0$, set $z_p = \psi_u(p)(Np)^{-s}$ and $t_p = z_p/(1+z_p)$. On $\operatorname{Re} s > 1$, deleting an additional surviving prime divides the squarefree Euler product by $1 + z_p$. Consequently the exact retained expression is

$$J_{u,U}(s) = F_{u,1}(s)B_{u,U}(s), \qquad B_{u,U}(s) = \sum_{d \text{ squarefree},\ (d,T_0)=1} \beta_U(d) \prod_{p \mid d} t_p.$$

Original zeros enforce $(d, u) = 1$ even when a sixth power drops from its primitive conductor. This factorization cannot restore a forbidden d row by replacing a zero sentinel with an unramified primitive value. The equality continues wherever both sides are holomorphic.

Without the cutoffs, $\beta_\infty(d) = \mu(d)2^{\omega(d)}$ on squarefree ideals. Absolute convergence justifies

$$B_{u,\infty} = \prod_{p \notin T_0} (1 - 2t_p) = \prod_{p \notin T_0} \frac{1 - z_p}{1 + z_p}, \qquad F_{u,1}B_{u,\infty} = \prod_{p \notin T_0} (1 - z_p) = \frac{1}{L^{T_0}(s, \psi_u)}.$$

Theorem 7 gives a quantitative rate rather than an unsupported exchange of this limit with a critical-strip contour.

A local counterexample to the naive square. In a one-prime projection at a good Eisenstein prime of norm 7, take the principal local phase and $s = 1$. Then $z = 1/7$, $t = 1/8$ and, for $U \geq 7$, the exact polynomial is $B_U = 1 - 2t = 3/4$. The unconstrained square $(1 - t)^2 = 49/64$ differs by $1/64$: it admits the forbidden pair $a = b = p$. After multiplying by $F = 1 + z = 8/7$ it gives $7/8$ instead of the correct $6/7$. This is an actual local Euler calculation, not an infinite-series or mean-square counterexample. It identifies the exact coprimality condition that a proposed mollifier-square simplification must preserve.

The exact intermediate coefficient band. For a squarefree m with $U < Nm \leq U^2$, one has $A_U(m) = -\mu(m)$. Indeed the unrestricted disjoint-assignment sum is $\mu(m)$. If $Na > U$, every $b \mid m/a$ automatically has $Nb \leq Nm/Na < U$, and $\sum_{b \mid m/a} \mu(b)$ vanishes unless $a = m$. Hence the excluded a assignments contribute exactly $\mu(m)$; the excluded b assignments contribute another $\mu(m)$. They cannot overlap because $NaNb > U^2 \geq Nm$ contradicts $ab \mid m$. Subtraction proves the claim. Together with the small-norm band, this gives $A_U = \mu$ for $Nm \leq U$ and $A_U = -\mu$ for $U < Nm \leq U^2$. The original smoothed columns have $Nm \geq D \geq U^2$, so the new band does not justify replacing their coefficients by $-\mu(m)$: generally they lie beyond it.

Theorem 8 (A stronger cutoff convergence would require zero-freeness)**.** *Fix a nonprincipal primitive ψ_u. If these actual $J_{u,U}$ converge locally uniformly on $\operatorname{Re} s > 1/2$ as $U \longrightarrow \infty$, then $L(s, \psi_u)$ has no zero there.*

Proof. Classical abelian Hecke continuation makes the numerator of $F_{u,1} = L^{T_0}(s, \psi_u)/L^{T_0}(2s, \psi_{u,2})$ holomorphic. Its denominator is nonzero on $\operatorname{Re} s > 1/2$ by the absolute Euler product at $2s$. Each $B_{u,U}$ is a finite sum with nonzero denominators $1 + z_p$ on $\operatorname{Re} s > 0$. Thus every $J_{u,U}$ is holomorphic on the asserted half-plane. A locally uniform limit G is holomorphic there. On $\operatorname{Re} s > 1$, the already proved convergence gives $L^{T_0}(s, \psi_u)G(s) = 1$. The identity theorem extends this equality to the connected half-plane, precluding a zero of L^{T_0}. The finitely deleted factors $1 - \psi_u(p)(Np)^{-s}$ do not vanish when $\operatorname{Re} s > 0$, so the same conclusion holds for the primitive L-function. □

This is an implication, not a proof that the stronger convergence holds, and it does not assert a converse. In particular, a family-averaged energy bound is not silently promoted to locally uniform convergence for each individual character. Principal rows have a pole at 1 and were excluded from this proposition. General Artin holomorphy is not an input.

Reproducible exact diagnostics. The new source `examples/arithmetic_cutoff/exact_recombination.py` uses actual split/inert prime ideals and raw sextic symbols in $\mathbb{Z}[\omega]$, with exact rational arithmetic. It checks the direct masked sum, coefficient regrouping, Euler packaging, both finite full-cutoff limits and original zeros, including third/sixth-power rows. Its overlapping- support negative controls are not tests of an infinite large sieve or of zero location. The analytic convergence argument above uses ordinary ideal Euler products and the explicitly stated Hecke continuation theorem; no new Lean declaration or complete GRH proof is claimed.

18 A cutoff-uniform envelope for the original masked remainder

Fix $K = \mathbb{Q}(\omega)$, $\mathcal{O} = \mathbb{Z}[\omega]$, a finite-order Hecke character ν, and a finite set S containing the primes above 6 and the conductor of ν. All unspecified ideal sums below are over nonzero integral ideals. Let $\chi_m(u) = (u/m)_6$ be the original symbol, zero when $(m,u) > 1$, and put

$$\beta_U(d) = \sum_{\substack{ab=d \\ Na, Nb \le U}} \mu(a)\mu(b), \qquad A_U(m) = \sum_{d \mid m} \beta_U(d).$$

The columns d, c, m are squarefree and prime to S where indicated. Retain the primitive characters $\psi_u = \mathrm{prim}(\nu\kappa_u)$, $\psi_{u,2} = \mathrm{prim}(\psi_u^2)$, and the full deleted set $T(u,d) = S \cup \mathrm{supp}(u) \cup \mathrm{supp}(d)$. For $H, D, U \ge 1$, $U^2 \le D$, and $W \in C_c^\infty((0,\infty))$ supported in $[1,2]$, the original normalized type-I piece is

$$I_u = D^{-1/2} \sum_{\substack{m \text{ squarefree} \\ (m,S)=1}} A_U(m)\nu(m)\chi_m(u) W(Nm/D). \tag{14}$$

Its exact principal residue is

$$P_u = \mathbf{1}_{\psi_u=1} \sqrt{D}\, \widehat{W}(1) \sum_{\substack{d \text{ squarefree} \\ (d,S)=1}} \frac{\beta_U(d)\nu(d)\chi_d(u)}{Nd} \rho_{T(u,d)},$$

$$\rho_T = \frac{\mathrm{res}_{s=1}\, \zeta_K(s)}{\zeta_K(2)} \prod_{p \in T} (1 + (Np)^{-1})^{-1}. \tag{15}$$

Define $E_u = I_u - P_u$; this is the same residue-subtracted remainder as the fully masked Mellin contour, with no cutoff limit or profile supremum.

Theorem 9 (Uniformity in the actual truncated cutoff)**.** *For every $\varepsilon > 0$,*

$$\sum_{0 < Nu \le H} |E_u|^2 \ll_{\nu,S,\varepsilon} \|W\|_\infty^2 (HD)^\varepsilon \{H + DH^{1/3} + (HD)^{2/3}\}. \tag{16}$$

The sum is over every actual nonzero Eisenstein element, including units, S-supported powers and sixth-power rows. The constant is independent of U under $U^2 \le D$. This is an application of the classical sextic large sieve with exact original coefficients, not a zero-location theorem.

Proof. On an absolutely convergent line, expand the restored squarefree series $F_{u,d}$. Its condition $(c, T(u,d)) = 1$ includes $(c,d) = 1$, and the original symbols multiply. Grouping $m = cd$ gives (14) by Mellin inversion. Each prime of m may belong to the first truncated Möbius divisor, to the second, or to neither. Consequently $|A_U(m)| \le 3^{\omega(m)} \ll_\eta (Nm)^\eta$, uniformly in U. The length-D coefficient sequence in (14) therefore has squared norm $\ll_\eta \|W\|_\infty^2 D^\eta$.

The all-ideal row norm in Section 3 of Blomer–Goldmakher–Louvel, arXiv:1112.1650v1 gives

$$B_2^1(X,D) \ll_\eta (XD)^\eta \{X + DX^{1/3} + (XD)^{2/3}\}.$$

For completeness, split $a = a_1 a_2^2 \cdots a_5^5 a_6^6$ dyadically; $a_1, \dots, a_5$ are squarefree and pairwise coprime. If their dyadic sizes are A_j, put $R = \prod_{j=1}^6 A_j$ and $M = \max(A_1, A_2)$. Pivot on the first or second variable, using the squarefree powered sieve proved simultaneously in that paper's Section 3.2. The other variables are bounded column multipliers. The block energy is at most

$$(XD)^\eta \{R + DR/M + D^{2/3}R/M^{1/3}\} \sum_m |\lambda_m|^2.$$

The inequalities $R \le X$, $R^3 \le XM^3$ and $R^3 \le X^2 M$ follow from $\prod_j A_j^j \le X$ and the pivot choice. Summing the logarithmically many blocks proves the displayed bound.

Finite S-unit and ray-class partitions convert the original literal symbol into the canonical column value $\vartheta_m(a)$ with fixed modulus-one coefficient multipliers. On noncoprime pairs both values are zero: squarefree m has outside-S conductor exactly m. This orientation preserves sixth-power zeros. At most six elements generate each outside-S ideal times an S-supported ideal; the S-supported sums converge at powers 1, 1/3 and 2/3. Thus (16) holds first with I_u in place of E_u.

A principal ψ_u requires every outside-S valuation of u to be divisible by 6. There are $O_{\nu,S}(H^{1/6})$ such literal rows. Equation (15), $|\rho_T| \ll_K 1$, and the actual coefficient sum

$$\sum_d \frac{|\beta_U(d)|}{Nd} \le \left(\sum_{Na \le U} \frac{1}{Na} \right)^2 \ll \log^2(2U)$$

give $\sum |P_u|^2 \ll DH^{1/6} \|W\|_\infty^2 \log^4(2U)$. This is absorbed by the right side of (16), using $U^2 \le D$ and a smaller internal exponent. Finally $|E_u|^2 \le 2|I_u|^2 + 2|P_u|^2$. The extended proof specifies the canonical-family transfer and the powered-sieve input precisely. □

Combining this with the previously proved fixed-twist AFM transfer yields

$$\sum_{0 < Nu \le H} |E_u|^2 \ll_{\nu,S,\delta,\varepsilon} \mathcal{M}_4(W)^2 (HD)^\varepsilon \min\{D^{2\delta} U^{2-4\delta} H,\ H + DH^{1/3} + (HD)^{2/3}\}. \tag{17}$$

At $H = D^{3/4}$ and $U = D^a$, $0 \le a \le 1/2$, the excess over H is at most $\min\{2\delta + (2-4\delta)a, 1/2\}$, up to arbitrarily small losses. The new envelope improves the old one for $a > 1/4$ as $\delta \downarrow 0$, but still leaves an exponent loss of 1/2. It does not settle the original signed short-row bound, type-II covariance, positive arithmetic determinant comparison or GRH. The principal-residue bound used here is sufficient for this envelope, not the stronger target needed to reassemble the full signed argument.

19 All-ideal reciprocity and the surviving block obstruction

The proof uses no new zero-free input. Its external analytic ingredient is the pinned BGL large sieve and the powered estimates established in its proof; elementary ideal counting and the already restored Mellin identity supply the remaining interfaces. All constants may depend on fixed K, S, ν and declared small exponents, never on U.

19.1 Exact recombination retains the primitive corrections

On $\operatorname{Re} s > 1$, start from the full defining series, with its original zero masks:

$$F_{u,d}(s) = \sum_{\substack{c \text{ squarefree} \\ (c,T(u,d))=1}} \nu(c)\chi_c(u)(Nc)^{-s}.$$

Its primitive ratio involves $L(s,\psi_u)$, the independently primitive $L(2s,\psi_{u,2})$ and every deleted Euler factor. Expanding this *restored* series gives the exact equality

$$\begin{aligned}\sum_{\substack{d \text{ squarefree} \\ (d,S)=1}} &\beta_U(d)\nu(d)\chi_d(u)(Nd)^{-s}F_{u,d}(s) \\ &= \sum_{\substack{m \text{ squarefree} \\ (m,S)=1}} A_U(m)\nu(m)\chi_m(u)(Nm)^{-s}. \end{aligned} \tag{18}$$

Indeed $(c,d)=1$ makes $m=cd$ squarefree, with $d \mid m$. If m meets u, both original symbols retain their zero. Only on the absolutely convergent line is this infinite series used. Mellin inversion there gives the finite smoothed polynomial (14). Nothing evaluates (18) as a Dirichlet series on $\operatorname{Re} s = 1/2+\delta$ or interchanges a cutoff limit with continuation across unknown zeros.

For squarefree m, a factorization $ab = d \mid m$ assigns each prime to a, to b, or to neither. The actual inequalities $Na, Nb \le U$ select a subset of these assignments. Thus

$$|A_U(m)| \le 3^{\omega(m)} \ll_\eta (Nm)^\eta.$$

The last divisor bound follows by choosing a fixed threshold so $3 \le (Np)^\eta$ beyond it, then absorbing the finitely many smaller prime ideals in a constant. When $Nm \le U$ all assignments occur, so $A_U(m) = \prod_{p\mid m}(1-1-1) = \mu(m)$; no such simplification is made for the length-D columns in general. With $\lambda_m = D^{-1/2}A_U(m)\nu(m)W(Nm/D)$, ideal counting and a smaller internal η give

$$\sum_m |\lambda_m|^2 \ll_\eta \|W\|_\infty^2 D^\eta, \qquad \operatorname{supp}\lambda \subseteq \{D \le Nm \le 2D\}. \tag{19}$$

19.2 The powered squarefree input and all-ideal pivot

Let ϑ_a be BGL's canonical primitive sixth-order family. Their Section 3 defines B_1^j using squarefree row/column ideals and kernel $\vartheta_a(m)^j$, while B_2^j uses *all* ideal rows and kernel $\vartheta_m(a)^j$. Theorem 1.3 states the squarefree bound for $j=1$; its Section 3.2 proves the estimates simultaneously for $1 \le j \le 5$, because the recursion in Lemma 3.4 uses $j\ell$. In particular the two inputs needed here are

$$B_1^j(M,D) \ll_\eta (MD)^\eta\{M + D + (MD)^{2/3}\}, \qquad j = 1,2. \tag{20}$$

To see the final step in the published recursion explicitly, its $E_{4/3}$ estimate is $M^{4/3} + D + (MD)^{2/3}$. If $M \le D$, its first term is at most $(MD)^{2/3}$; if $M > D$, use reciprocity/duality (3.2) and the estimate with M, D reversed. The finite set of powers is treated together; no inference from a fixed conductor count to an arbitrary primitive-character sieve occurs.

Write each ideal row uniquely as

$$a = a_1 a_2^2 a_3^3 a_4^4 a_5^5 a_6^6,$$

where $a_1, \dots, a_5$ are pairwise-coprime squarefree ideals and a_6 is arbitrary. For valuation $6q+r$, $0 \le r \le 5$, place the prime in a_r if $r > 0$ and its power q in a_6. Dyadically let $A_j \le Na_j < 2A_j$,

with $A_j \ge 1$, and put $R = \prod_{j=1}^{6} A_j$. A nonempty block satisfies $\prod_j A_j^j \le X$ when $Na \le X$. Choose $\ell = 1$ if $A_1 \ge A_2$, and $\ell = 2$ otherwise; put $M = A_\ell$. Keep every other ideal fixed. Evaluation of the single *column* character gives

$$\vartheta_m(a) = \vartheta_m(a_\ell)^\ell \vartheta_m(a/a_\ell^\ell).$$

The second factor is a bounded column multiplier, including its original zero if m meets any fixed factor. It may depend on all the fixed ideals but not on the pivot row. Pairwise-coprimality and the exact row norm restriction merely select a subset of the positive pivot-row sum. Apply (20) to the first factor. There are $O_K(R/M)$ choices of the other ideals by ideal counting. The block energy is bounded by

$$(XD)^\eta \{R + DR/M + D^{2/3}R/M^{1/3}\} \sum_m |\lambda_m|^2. \tag{21}$$

All three monomial bounds used to sum these blocks are elementary. First $R \le \prod_j A_j^j \le X$. If $M = A_1$, then

$$(R/M)^3 = A_2^3 \cdots A_6^3 \le A_1 A_2^2 A_3^3 A_4^4 A_5^5 A_6^6 \le X,$$

since $A_2 \le A_1$ and $j \ge 3$ for the remaining factors. If $M = A_2$, use $A_1 \le A_2$ in the same comparison. Finally,

$$\frac{R^3}{M(\prod_j A_j^j)^2} = \frac{A_1}{M A_2 A_3^3 A_4^5 A_5^7 A_6^9} \le 1.$$

Thus $R/M \le X^{1/3}$ and $R/M^{1/3} \le X^{2/3}$. Insert these in (21) and sum the logarithmically many blocks. This proves the all-ideal estimate used in Theorem 9. It is precisely the mechanism of BGL Lemma 3.4 and (3.6)–(3.9), with the final powered bound inserted. The simpler fixed-squarefull-row argument would leave the weaker $D\sqrt{X}$ term.

19.3 Literal rows, reciprocal orientation and principal count

Let $a = (u)_S$ denote the outside-S ideal. The finite S-unit comparison gives $\kappa_u = \eta_u \vartheta_a$ as primitive characters, with η_u in a finite fixed set. For $(a, m) = 1$, BGL reciprocity therefore gives

$$\chi_m(u) = \eta_u(m) C([a], [m]) \vartheta_m(a). \tag{22}$$

The notation is deliberate: the left side is the project's original $(u/m)_6$, not BGL's canonical character indexed by m. When $(a, m) > 1$, the left side is zero. The right side is also zero, because squarefree m has outside-S primitive conductor exactly m. Equation (22) hence extends to every pair. Using $\vartheta_a(m)$ on noncoprime pairs instead would be false: primitive ϑ_{b^6} has restored primes of b, whereas the original symbol must still vanish there.

Partition rows by η_u and $[a]$, both finite. On each part the factor $\eta_u(m)C([a], [m])$ is a fixed modulus-one column multiplier. A positive row sum over that part is bounded by the full B_2^1 row sum. For each S-supported ideal b and outside-S ideal a, at most six actual elements generate ab. Thus the required literal energy is bounded by the all-ideal norm summed over b, with $X = H/Nb$. The geometric products

$$\sum_{b \ S\text{-supported}} (Nb)^{-q} = \prod_{p \in S} (1 - (Np)^{-q})^{-1} < \infty, \qquad q = 1,\ 1/3,\ 2/3,$$

transfer the three terms without losing any unit or S-power row. Together with (19), this proves the claimed I energy.

If ψ_u is principal, ν is unramified outside S, and tame Kummer inertia at $v_p(u)$ has order $6/\gcd(6, v_p(u))$. It must be trivial, so each such valuation is divisible by 6. Writing $(u) = ba_6^6$ with b supported in S, ideal counting gives

$$\#\{0 < Nu \le H : \psi_u = 1\} \ll_K H^{1/6} \prod_{p \in S} (1 - (Np)^{-1/6})^{-1}.$$

This is only a necessary condition, sufficient for an upper count. The exact residue formula (15) and $|\widehat{W}(1)| \le \|W\|_\infty$ then give the principal energy bound used above. No sign or positivity of that residue is assumed.

19.4 What the envelope still fails to prove

The $DH^{1/3}$ term is still larger than $H(HD)^\varepsilon$ at $H = D^{3/4}$. The monomial comparison itself is saturated by $A_1 = A_2 = X^{1/3}$ and $A_3 = \cdots = A_6 = 1$. This proves sharpness of that *particular comparison*, not an energy lower bound for actual arithmetic coefficients. In particular, no conclusion that the signed A_U cannot cancel is justified. The next load-bearing input must exploit those actual truncated Möbius coefficients inside the mixed moment, or establish the required signed type-II minus type-I covariance. The smaller-scale short term, principal contribution, full arithmetic determinant identification and GRH remain separate obligations.

Exact finite checks use the split Eisenstein primes generated by $3 + \omega$, $4 + \omega$, $5 + 2\omega$, $6 + \omega$, of norms $7, 13, 19, 31$. Their actual residue symbols are computed in $\mathbb{Q}(\zeta_6)$, retaining sixth-power zeros. Coefficient recombination and dyadic inequalities pass; they check finite algebra and bookkeeping, not the external analytic sieve or the contour theorem. No new Lean declarations accompany this argument.

20 High cutoffs, actual bilinear factors and a signed covariance test

Retain the original Eisenstein family, fixed finite-order ν, fixed S, common $W \in C_c^\infty((0,\infty))$ supported in $[1, 2]$, and every raw shared-prime zero. Put

$$C_U(m) = \sum_{\substack{abc=m \\ Na>U,\, Nb>U}} \mu(a)\mu(b), \qquad r_U(m) = \#\{(a, b) : ab = m,\ Na, Nb > U\}.$$

All factors are ideals, squarefree and pairwise coprime; on the active annulus $A_U = C_U - \mu$. A coefficient identity is different from a row-energy bound.

Proposition 10 (The high-cutoff window)**.** *If $D > 1$ and $2D/7 \le U^2 \le D$, then for every surviving squarefree column $D \le Nm \le 2D$,*

$$C_U(m) = \mu(m) r_U(m), \qquad A_U(m) = \mu(m)(r_U(m) - 1), \qquad |A_U(m)| \ge 1.$$

For each fixed nonzero W, the normalized squared coefficient norm satisfies

$$D^{-1} \sum_{\substack{m \text{ sf} \\ (m,S)=1}} |A_U(m) W(Nm/D)|^2 \gg_{W,S} 1 \tag{23}$$

for large D, uniformly in this window. This is not a lower bound for row energy.

Proof. The least norm of a nonunit ideal outside S is at least 7. A long-long term with $c \neq 1$ would have $Nm > 7U^2 \geq 2D$. Thus $ab = m$, including both endpoints. The fixed-point-free swap $(a, b) \mapsto (b, a)$ makes r_U even, so A_U is an odd nonzero integer. The last claim follows by counting squarefree ideals on a fixed interval where $|W|$ is bounded below; the constant and error are proved in the extended proof. □

The exact remaining polynomial is genuinely bilinear, with $(a, b) = 1$, $U < Na, Nb < 2D/U$, $D \leq Na\,Nb \leq 2D$, and $x_{ab}(u) = x_a(u)x_b(u)$, including zeros. Both factors see the same row $u = \varepsilon\alpha\beta^2$. Cauchy followed by the accepted powered BGL bound gives only

$$Q_{B_X}(C_U, C_U) \ll_{\nu,S,\epsilon} (DX)^{\epsilon}\|W\|_{\infty}^2\{\sqrt{D}\,X^2 + DX + D^{5/6}X^{5/3}\}. \tag{24}$$

At $X \asymp D^{1/4}$ the exponents are $1, 5/4, 5/4$; the earlier regrouped bound has $1/2, 5/4, 13/12$ and is stronger. No BT_σ saving is obtained. Multiplying two marginal operator norms does not supply a coupled fourth moment.

A finite test in the original family. There is no automatic nonnegative signed covariance, even when $C_U(m)\mu(m) \geq 0$. Take $\nu = 1$, S the primes above $2, 3$, $D = 400$, $U = 17$, and all 72 literal rows of B_7. Choose the common nonnegative smooth profile constructed in the extended proof, with values 10 at 439/400, 1 at 475/400, and zero at every other integer norm divided by 400. There are exactly four active squarefree ideal columns, of norms $439, 439, 475, 475$. In that order,

$$A = (1, 1, 1, 1), \quad C = (0, 0, 2, 2), \quad \mu = (-1, -1, 1, 1), \qquad \Re G_{B_7} = \begin{pmatrix} 72 & 0 & 0 & 18 \\ 0 & 72 & 18 & 0 \\ 0 & 18 & 72 & 0 \\ 18 & 0 & 0 & 72 \end{pmatrix}.$$

Exact raw-symbol enumeration, independently reproducible from the ideal and residue-field definitions, gives

$$Q(C, \mu) = -\frac{27}{25}, \qquad Q(C, C) = \frac{36}{25}, \quad Q(\mu, \mu) = \frac{864}{25}, \quad Q(A, A) = \frac{954}{25}. \tag{25}$$

The real Gram matrix is positive definite, with eigenvalues $54, 90$; its positivity does not force a cross covariance to be positive. This fixed finite example refutes that shortcut, not the asymptotic BT_σ target or the original signed mean-square estimate.

The two-cutoff contour remains coupled. For $T_0 = S \cup \mathrm{supp}(u)$, $\psi = \psi_u$ and independently primitive $\psi_2 = \mathrm{prim}(\psi^2)$, put $q_p = \psi(p)(Np)^{-s}$, $\alpha_p = (Np)^{-z}$, $\beta_p = (Np)^{-w}$ outside T_0. The actual three-placement Euler factor is $1 + q_p - q_p\alpha_p - q_p\beta_p$. Its exact factorization is

$$\mathcal{F}(s, z, w) = \frac{L^{T_0}(s, \psi)}{L^{T_0}(s + z, \psi)L^{T_0}(s + w, \psi)} \prod_{p \notin T_0} \left(1 - \frac{q_p^2(1 - \alpha_p)(1 - \beta_p)}{(1 - q_p\alpha_p)(1 - q_p\beta_p)}\right). \tag{26}$$

Exact double Perron inversion is valid on positive auxiliary lines $c, d > 1 - \Re s$ for $1/2 < \Re s < 1$, retaining the original principal pole exclusion. These safe lines reproduce the previous $U^{2-4\delta}$ energy loss at $\Re s = 1/2 + \delta$, with the same common-profile derivative and fixed-margin dependencies. The correction product alone does not justify moving those lines through reciprocal-L poles. In fact $\mathcal{F}(s, z, 0) = 1/L^{T_0}(s + z, \psi)$ as a meromorphic identity. No zero-free region, reciprocal moment, analytic saving, new Lean formalization or GRH proof is supplied by this construction.

21 Proof details for the high-window and two-cutoff tests

21.1 The coefficient lower bound and its precise limitation

The element lattice $\mathbb{Z}[\omega] \subset \mathbb{C}$ has covolume $\sqrt{3}/2$. Counting disks with boundary error $O(\sqrt{x})$ and dividing by the six units, using that this ring is a PID, gives

$$N_K(x) = \kappa x + O(\sqrt{x}), \qquad \kappa = \frac{\pi}{3\sqrt{3}}.$$

Finite inclusion-exclusion at S gives $N_{K,S}(y) = \kappa y \prod_{p\in S}(1-(Np)^{-1}) + O_S(\sqrt{y})$. Insert $1_{\mathrm{sf}}(m) = \sum_{d^2\mid m}\mu(d)$. Absolute convergence at 2, the tail $\sum_{Nd>\sqrt{x}}(Nd)^{-2} = O(x^{-1/2})$, and $\sum_{Nd\le\sqrt{x}}(Nd)^{-1} = O(\log(2x))$ yield

$$N_{\mathrm{sf},S}(x) = c_S x + O_S(\sqrt{x}\log(2x)), \qquad c_S = \frac{\kappa}{\zeta_K(2)}\prod_{p\in S}(1+(Np)^{-1})^{-1} > 0.$$

These tail estimates follow by partial summation from the displayed lattice count. For a fixed nonzero W, choose $[a,b] \subset (1,2)$ with $|W| \ge \eta > 0$. Proposition 10 gives, for large D,

$$D^{-1}\sum |A_U(m)W(Nm/D)|^2 \ge \tfrac{1}{2}\eta^2 c_S(b-a).$$

There is no uniform lower constant over all profiles with one specified supremum norm. Nor can a coefficient lower bound be combined with an operator upper bound to produce a row-energy lower bound: no lower singular value has been established on the actual vector.

21.2 What legal factor Cauchy actually proves

For dyadic $Na \asymp A, Nb \asymp B$, retain the exact $(a,b)=1$ mask, hyperbolic support and $W(Na\,Nb/D)$ in

$$T_a(u) = \sum_b \mu(b)\nu(b)W(Na\,Nb/D)x_b(u).$$

Cauchy gives

$$|\sum_a \mu(a)\nu(a)x_a(u)T_a(u)|^2 \ll A\sum_a |T_a(u)|^2.$$

For fixed a, fixed second row factor and fixed unit/ray classes, the b coefficients are independent of the pivot row and have squared norm $\ll B\|W\|_\infty^2$. The prior classical power-one/two BGL input has $\Delta(X,B) = X + B + (XB)^{2/3}$. Summing the $O(A)$ fixed columns and $O(X)$ other row factors gives

$$\mathcal{E}_{A,B} \ll_{\nu,S,\epsilon} (DX)^\epsilon \|W\|_\infty^2 \frac{A^2B}{D} X\Delta(X,B) \ll (DX)^\epsilon\|W\|_\infty^2\{AX^2 + DX + D^{2/3}A^{1/3}X^{5/3}\}.$$

In the high window $A, B \asymp \sqrt{D}$, and only boundedly many dyadic pairs occur. This proves (24); exchanging factors or row pivots leaves its DX term. A product of marginal second-moment bounds supplies no estimate for the two transforms on the same (r,s). Any useful saving must control that coupled fourth moment or the signed actual-coefficient covariance, with all masks retained.

21.3 A smooth profile and the exact covariance computation

Let

$$\phi(t)=\begin{cases}\exp(1-1/(1-t^2)), & |t|<1,\\ 0, & |t|\ge 1,\end{cases}\qquad W_x(y)=x\phi(4(400y-439))+\phi(4(400y-475)).$$

For $x\ge 0$ this is nonnegative and smooth; its disjoint supports lie strictly inside $(1,2)$. For every integer n, $W_x(n/400)$ equals x at $n=439$, 1 at $n=475$ and zero otherwise. Thus the computation uses exact values of a genuine common C_c^∞ profile, rather than claiming a finite rational weight is smooth. At $x=10$, $\|W_x\|_\infty=10$.

The primary ideal generators, in the order used above, are

$$(-23,-18),\quad (-5,18),\quad (10,-15),\quad (25,15),$$

where (a,b) denotes $a+b\omega$. The first two have norm 439. The last two factor as $(-2,3)(-5,0)$ and $(-5,-3)(-5,0)$, with norms $19\cdot 25$. Since $19,25>17$, the long-factor count is 2 there and zero on the prime columns. All squarefree ideals at both selected norms are included. The complete B_7 uses the four good primes of norms $7,7,13,13$ for distinct ordered row factors and all six units. It has 72 distinct rows, maximum norm 2197; each is nonprincipal by its outside-S Kummer valuation 1 or 2 modulo 6, whose nontrivial tame character has conductor exponent 1.

For the raw Gram $G_{ij}=\sum_{u\in B_7} x_{m_i}(u)\overline{x_{m_j}(u)}$, exact finite-field exponentiation and lattice enumeration give the displayed real matrix. The weighted real vectors are $\ell=(0,0,2,2)$, $v=(-x,-x,1,1)$, and $a=(x,x,1,1)$. Therefore

$$Q(C_U,\mu)=D^{-1}\ell^t(\Re G)v=\frac{36-9x}{50}.$$

Its diagonal contribution is $36/50>0$ and its off-diagonal contribution is $-9x/50$. For $x=10$ this is $-27/25$ and gives (25). The identity $(\Re G-72I)^2=18^2I$ proves the real eigenvalues $54,90$; these are not being asserted as the eigenvalues of the full complex Gram. The exact source is `examples/arithmetic_cutoff/signed_covariance.py`. A second test retains raw zeros on the 246 rows $0<Nu\le 89$ whose outside-S part is nonunit. Its covariance at $x=10$ is $-69/50$. This subset is guaranteed nonprincipal; it is not called the entire nonprincipal family. Here $89=\lfloor 400^{3/4}\rfloor$. These are finite counterexamples to automatic covariance positivity, not asymptotic counterexamples to BT_σ or GRH.

21.4 Double Perron without a divergent critical-strip series

For $T_0=S\cup\operatorname{supp}(u)$ retain $F_0(s)=L^{T_0}(s,\psi)/L^{T_0}(2s,\psi_2)$, with the independently primitive square character. Set $t_p=q_p/(1+q_p)$ outside T_0. The finite actual cutoff is

$$J_U(s,u)=F_0(s)B_U(s),\qquad B_U(s)=\sum_{\substack{a,b\ \mathrm{sf},\ (a,b)=1\\ (ab,T_0)=1,\ Na,Nb\le U}}\mu(a)\mu(b)\prod_{p\mid ab}t_p.$$

For $\sigma=\Re s>1/2$, the series

$$\mathcal{G}(s,z,w)=\sum_{\substack{a,b\ \mathrm{sf},\ (a,b)=1\\ (ab,T_0)=1}}\mu(a)\mu(b)\prod_{p\mid ab}t_p\,(Na)^{-z}(Nb)^{-w}=\prod_{p\notin T_0}\left(1-t_p(Np)^{-z}-t_p(Np)^{-w}\right)$$

converges absolutely on $\sigma+\Re z>1$, $\sigma+\Re w>1$, uniformly in all vertical parts at fixed positive margins. This follows from $|t_p|\le (Np)^{-\sigma}/(1-(Np)^{-\sigma})$. There $\mathcal{F}=F_0\mathcal{G}$, without evaluating an infinite c series in the critical strip. Initial absolute convergence also proves (26) by the local identity

$$(1-q)(1+q-q\alpha-q\beta)=(1-q\alpha)(1-q\beta)-q^2(1-\alpha)(1-\beta).$$

The correction product is holomorphic on

$$\sigma+\Re z>0,\quad \sigma+\Re w>0,\quad 2\sigma+\min(\Re z,0)+\min(\Re w,0)>1.$$

On margins $\sigma+\Re z,\sigma+\Re w\ge\eta>0$ and last exponent $\ge 1+\epsilon$, its local deviation from 1 is at most $4(1-2^{-\eta})^{-2}(Np)^{-1-\epsilon}$. This proves normal convergence and a phase/deletion-uniform vertical upper bound. Nonvanishing of the correction is not asserted.

For $1/2<\sigma<1$, positive lines $c,d>1-\sigma$, and $V=\lfloor U\rfloor+1/2$, define

$$\mathcal{I}_{T_1,T_2}=\frac{1}{(2\pi i)^2}\int_{c-iT_1}^{c+iT_1}\int_{d-iT_2}^{d+iT_2}\mathcal{F}(s,z,w)\frac{V^{z+w}}{zw}\,dw\,dz.$$

The half-integer V includes the actual equality cutoff, since $Na\le U$ is exactly $Na<V$. Put $\Delta_V=\inf_{n\in\mathbb{Z}_{\ge1}}|\log(V/n)|\ge 1/(2V+1)$, and let $\mathcal{A}=\sum_{a,b}|\mu(a)\mu(b)\prod_{p\mid ab}t_p|(Na)^{-c}(Nb)^{-d}$ with the same disjoint support. For $c,T>0$ and $v>0$, $v\ne1$, define

$$K_{c,T}(v)=\frac{1}{2\pi i}\int_{c-iT}^{c+iT}\frac{v^z}{z}\,dz.$$

Finite-height termwise integration of the absolutely convergent $\mathcal{G}$ and the one-variable tail estimate $|K_{c,T}(v)-1_{v>1}|\le 2v^c/(\pi T|\log v|)$ give

$$|\mathcal{I}_{T_1,T_2}-J_U|\le|F_0|\mathcal{A}V^{c+d}\left\{\frac{2}{\pi T_1\Delta_V}+\frac{2}{\pi T_2\Delta_V}+\frac{4}{\pi^2T_1T_2\Delta_V^2}\right\}. \tag{27}$$

Thus $\mathcal{I}_{T_1,T_2}\to J_U$ as both heights tend independently to infinity, away from the original principal pole. This is an oscillatory limit, not an absolutely integrable infinite vertical integral.

Taking $T_1=T_2=V^2$ gives the safe upper bound $|J_U|\ll(Nu)^\eta|L(s,\psi)|V^{c+d}\log^2(2V)$, with constants retaining fixed margins, ν,S, and σ. Choosing c,d close to $1-\sigma$ reproduces $U^{2-2\sigma+\eta}$ in amplitude and $U^{2-4\delta+\eta}$ in energy on $\sigma=1/2+\delta$. Insertion into the original common-profile Mellin argument retains $M_4(W)^2$ and all previously required height/conductor uniformities. No derivative-free improvement follows from this Perron estimate.

The correction converges even on some negative auxiliary lines, but moving there can cross $z=\rho-s$ or $w=\rho-s$, from zeros of the primitive L-function. The exact meromorphic slice $\mathcal{F}(s,z,0)=1/L^{T_0}(s+z,\psi)$ forbids assuming universal cancellation of these poles. At a numerator zero the slice must be understood meromorphically, not by an unjustified pointwise substitution. Sequential shifts have single boundary integrals and crossed-zero residues as well as the kernel double residue; away from s zeros or poles that double residue is $1/L^{T_0}(s,\psi)$. A principal L pole supplies a reciprocal zero, not a reciprocal pole. Uniform control of actual reciprocal integrals/residues or the signed covariance remains a new input. Numerator AFM alone supplies neither.

22 A quantitative covariance budget without a new exponent

Retain the original Eisenstein sextic symbols, fixed finite-order ν, fixed S, one possibly complex common $W\in C_c^\infty$ supported in $[1,2]$, all squarefree columns prime to S, and every raw shared-prime

zero. In the accepted window $2D/7 \le U^2 \le D$, put $r_m = r_U(m)$, $\lambda_m = \mu(m)\nu(m)W(Nm/D)$ and $v_m = \overline{\lambda_m}$. Thus $C_U = \mu r$ and $A_U = \mu(r-1)$ on the active annulus. For a finite literal row block B, let

$$G_{mn} = \sum_{u\in B} x_m(u)\overline{x_n(u)}, \qquad R = \operatorname{diag}(r_m), \qquad M_j = \sum_m r_m^j |W(Nm/D)|^2 \quad (0 \le j \le 2).$$

The primitive $\psi_u, \psi_{u,2}$ and full deleted sets retain their previous meanings; no conductor-drop value replaces a raw zero.

Proposition 11 (An actual-vector lower budget)**.** *If $0 \le G \le \Lambda_B I$, then*

$$\Re Q_B(C_U, \mu) \ge -\frac{\Lambda_B}{2D}\{\sqrt{M_0 M_2} - M_1\}. \tag{28}$$

In this high window, uniformly in U,

$$\sum_{D\le Nm\le 2D} r_m \ll_K D, \qquad \sum_{D\le Nm\le 2D} r_m^2 \ll_K D\log(2D). \tag{29}$$

Consequently

$$\Re Q_B(C_U, \mu) \ge -C_K \|W\|_\infty^2 \Lambda_B \sqrt{\log(2D)}.$$

The vector inequality is sharp among positive contractions with the specified v, Rv; no such sharpness is asserted for the arithmetic Gram. The moment bound counts actual balanced ideal factorizations, rather than assuming independence of the two row factors.

On B_X, inserting the previously accepted powered-BGL norm gives only

$$\Re Q_{B_X}(C_U, \mu) \ge -C_{\nu,S,\epsilon}\|W\|_\infty^2 (DX)^\epsilon \{X^2 + DX + D^{2/3}X^{5/3}\}.$$

At $X \asymp D^{1/4}$ the exponents remain $1/2, 5/4, 13/12$. At $H = D^{3/4}$ this budget still has the old $D^{1/2}$ excess over H; it supplies no BT_σ saving. A negative-part budget is weaker than $2\Re Q(C_U, \mu) \ge Q(C_U, C_U) + Q(\mu, \mu) - R_\sigma$.

For $S_R = \sqrt{R}$, the exact identity

$$\tfrac{1}{2}(RG + GR) = S_R G S_R + \tfrac{1}{2}[S_R, [S_R, G]]$$

localizes the commutator to unequal factor-count levels, with entry $(\sqrt{r_m} - \sqrt{r_n})^2 G_{mn}$. If B retains all six literal units, the actual unit sum further gives $G_{mn} = 0$ unless $Nm \equiv Nn \pmod{36}$. These are exact deletions, not a power saving. Even zero different-level pairings leave the nonnegative within-level energies $\sum_k (k-1)^2 Q_B(\mu \mathbf{1}_{r=k}, \mu \mathbf{1}_{r=k})$. The unequal-product coupled fourth moment and its mixed Möbius term remain unestimated. No new analytic exponent, energy lower bound, Lean formalization or GRH conclusion follows.

23 Proof and arithmetic boundary of the covariance budget

23.1 The positive-contraction calculation

Since G is the exact raw Gram, it is positive semidefinite. If $\Lambda_B > 0$, put $T = 2G/\Lambda_B - I$, a self-adjoint contraction. Then

$$\Re(v^* R G v) = \frac{\Lambda_B}{2}\{M_1 + \Re(v^* R T v)\} \ge \frac{\Lambda_B}{2}\{M_1 - \|Rv\|_2 \|v\|_2\},$$

which is (28); $\Lambda_B = 0$ is immediate. The Hermitian rank-two matrix $\{(Rv)v^* + v(Rv)^*\}/2$ has possible nonzero eigenvalues $(M_1 \pm \sqrt{M_0 M_2})/2$. Its negative spectral projection realizes the lower bound among all $0 \le G \le \Lambda_B I$, a linear-algebra sharpness statement only. If $a \le r_m \le b$ on the weighted support, $a + b > 0$, the pointwise inequality $r_m^2 \le (a+b)r_m - ab$ also gives

$$\Re Q_B(C_U, \mu) \ge -\frac{\Lambda_B M_0 (b-a)^2}{8D(a+b)}.$$

For $a = b = 0$ the covariance is zero. The constant follows by maximizing $\sqrt{(a+b)t - ab} - t$ on $[a, b]$.

23.2 The balanced-divisor second moment

Set $T_0 = \sqrt{D}$. The high-window factor restrictions imply

$$\sqrt{2/7}\, T_0 < Na, Nb < \sqrt{14}\, T_0.$$

The elementary ideal count $\#\{a : Na \le Y\} \ll_K Y$ proves the first assertion of (29). For the second, two ordered factorizations $ab = cd = m$ have the unique form

$$a = gh, \quad c = gk, \quad b = k\ell, \quad d = h\ell, \qquad g = (a, c).$$

Squarefreeness makes g, h, k, ℓ pairwise coprime and squarefree. To upper-bound this positive count, enlarge to arbitrary ideals. For $M = \max(Nh, Nk)$ the individual bounds are

$$Ng \le \frac{\sqrt{14}\, T_0}{M}, \qquad N\ell \le \frac{\sqrt{14}\, T_0}{M}, \qquad 1 \le M \le \sqrt{14}\, T_0.$$

On $2^j \le M < 2^{j+1}$ there are $O_K(2^{2j})$ choices of (h, k) and $O_K((T_0/2^j)^2)$ choices of (g, ℓ), with bounded terminal ranges absorbed in the constant. Each dyadic range contributes $O_K(D)$; there are $O(\log(2D))$ ranges. This proves the second moment with no extra logarithm from counting only $NgN\ell$. Restricting to primes outside fixed S decreases the count. Hence

$$M_0, M_1 \ll_K \|W\|_\infty^2 D, \qquad M_2 \ll_K \|W\|_\infty^2 D \log(2D).$$

These constants are independent of the cutoff in the stated window. Only the common supremum norm is used, including for complex W.

23.3 What the commutator actually controls

Write

$$t_{mn} = D^{-1} \Re\{\lambda_m \overline{\lambda_n} G_{mn}\}, \qquad \delta_{mn} = (\sqrt{r_m} - \sqrt{r_n})^2.$$

The exact commutator identity proves

$$\Re Q_B(C_U, \mu) \ge Q_B(\sqrt{r}\,\mu, \sqrt{r}\,\mu) - \tfrac{1}{2} \sum_{m,n} \delta_{mn} |t_{mn}|.$$

The displayed loss retains the actual character and profile phases before taking absolute values; it is at most the phase-blind loss obtained from $|t_{mn}| \le D^{-1} |\lambda_m \lambda_n G_{mn}|$. On the accepted four-column smooth B_7 fixture at amplitude $x \ge 0$, the positive term is $18/25$, this coefficient-adapted loss is $9x/50$, and the lower bound is attained. Purely imaginary entries of the full complex Gram make the phase-blind loss larger. This calibrates the new budget and does not give an asymptotic obstruction.

For the unit deletion let $\zeta_6 = 1+\omega$. Every good prime norm q is $1 \pmod 6$, and direct residue-field exponentiation gives $x_p(\zeta_6) = \zeta_6^{(q-1)/6}$. Multiplying over $p \mid m$ and reducing $Nm = \prod_{p|m}(1+6e_p)$ modulo 36 gives

$$x_m(\zeta_6) = \zeta_6^{(Nm-1)/6 \bmod 6}.$$

Multiplication of all literal rows by ζ_6 preserves their raw zeros and proves the norm-congruence deletion in the core. The vector budget may be applied separately on these six classes. No D-dependent number of classes or conductor saving is produced.

The moment proof above counts the coefficient diagonal $ab = cd$. It has not estimated the unequal-product character quadruples in the coupled row fourth moment. Replacing the inherited DX norm by $D^{1-\sigma}X$ merely because equal-level and different-class entries vanish is the unsupported step. Small negative commutator loss alone also leaves the within-level energies in the core. Principal residues are unchanged; this covariance is used on the nonprincipal blocks where the original residue is zero, not as a new centering prescription.

23.4 The complete near-saturating face

For a six-factor row block let $R_0 = \prod_{j=1}^{6} A_j$, $M = \max(A_1, A_2)$ and $\prod_j A_j^j \le H$. The accepted exact-mask block norm is

$$\Lambda_A \ll_{\nu,S,\epsilon} (HD)^{\epsilon}\{R_0 + DR_0/M + D^{2/3}R_0/M^{1/3}\}.$$

The column moment and covariance proofs apply to every such block. Write $x_j = \log A_j$, $h = \log H$, $c_0 = \sum_j jx_j$ and $r_0 = \sum_j x_j$. Then, for $x_1 \ge x_2$,

$$h - 3(r_0 - \max(x_1, x_2)) = h - c_0 + (x_1 - x_2) + x_4 + 2x_5 + 3x_6;$$

for $x_2 \ge x_1$ replace $x_1 - x_2$ by $2(x_2 - x_1)$. All terms are nonnegative. Thus the middle monomial is saturated on

$$A_1 = A_2, \qquad A_1A_3 = H^{1/3}, \qquad A_4 = A_5 = A_6 = 1,$$

with full capacity. The balanced $A_3 = 1$ case is only an endpoint. The a_3 direction remains nonprincipal by its outside-S valuation $3 \pmod 6$. If $R_0/M \ge H^{1/3}D^{-\eta}$, the total displayed slack is at most $3\eta \log D$. Therefore a fixed power gain must be uniform on this full neighboring region. The proved budget still costs DR_0/M; it gives no global improvement over the accepted envelope.

The independent standard-library diagnostic checks nine ideal moment cases, eight raw Gram cases with 27 000 phase values and 5 184 raw zeros, exact radical commutator/vector budgets, and 227 exponent cases. Finite tables use common norm profiles realizable by smooth disjoint bumps; they are not tests of an infinite analytic estimate. Full derivation, source bindings and preserved checks are under `planning/covariance-budget-20261105/attempt/` and `outputs/checks/2026-11-05-covariance-budget/attempt/`.

24 The fourth-moment conductor and its full raw zero set

Retain the original Eisenstein sextic symbols $x_m(u) = (u/m)_6$, fixed finite-order ν, fixed S, and one possibly complex common smooth W supported in $[1,2]$. Put $\lambda_m = \mu(m)\nu(m)W(Nm/D)$ and $r_m = r_U(m)$. Columns remain squarefree and prime to S. In $2D/7 \le U^2 \le D$, the actual polynomial is

$$\mathcal{C}_U(u) = \sum_{\substack{(a,b)=1,\ Na,Nb>U \\ D\le Na\,Nb\le 2D}} \mu(a)\mu(b)\nu(ab)W(Nab/D)x_a(u)x_b(u).$$

Its squared row energy has four factor indices a, b, c, d, with $(a, b) = (c, d) = 1$. These internal masks are essential. At each prime,

$$e_p = \mathbf{1}_{p|a} + \mathbf{1}_{p|b} - \mathbf{1}_{p|c} - \mathbf{1}_{p|d} \quad \text{belongs to } \{0, \pm 1\}.$$

Thus, for the two squarefree products $m = ab, n = cd$, the primitive oscillatory conductor outside fixed S is $q = mn/(m, n)^2$. The full raw zero-preserving period is $L = mn/(m, n)$:

$$x_m(u)\overline{x_n(u)} = \mathbf{1}_{(u,(m,n))=1}\theta_{m/(m,n),\, n/(m,n)}(u).$$

No common-prime zero is restored by taking the primitive character.

A generic fourth moment $|\sum_m \lambda_m x_m(u)|^4$ has four individually squarefree columns $m_1, \dots, m_4$ with no such internal coprimality. Its signed local exponent $e_p = \sum_{i=1}^{2} \mathbf{1}_{p|m_i} - \sum_{i=3}^{4} \mathbf{1}_{p|m_i}$ can also be ± 2, giving a nontrivial cubic component. For $L = \mathrm{rad}(m_1 m_2 m_3 m_4)$, the exact row-power interface is

$$Q_j = \prod_{\substack{p|L \\ je_p \not\equiv 0 \ (6)}} p, \qquad Z_j = L/Q_j, \qquad \prod_{i=1}^{2} x_{m_i}(v^j)\overline{\prod_{i=3}^{4} x_{m_i}(v^j)} = \theta_{j,e}(v)\mathbf{1}_{(v,Z_j)=1}.$$

Here $\theta_{j,e}$ is primitive on Q_j, with its own zero extension. At $j = 3$ an exponent ± 2 becomes primitive-principal but its raw zero remains. For the original two-squarefree-product expansion, $Q_j = q$ for all $1 \le j \le 5$, including the quadratic $j = 3$ direction. For the accepted row blocks $Nr_j \asymp A_j$, put $R_0 = \prod_j A_j$, $P = \max(A_1, \dots, A_5)$ and retain $\prod_j A_j^j \le H$. The full A_3 saturation face and all row-factor masks remain.

The small-conductor pair count recovers an already controlled high-gcd sector; it is not a new analytic advance. On the remaining original sector $Nq > P^2$, favorable smooth completion costs $\min(A_j, \sqrt{Nq}) = A_j$ for $1 \le j \le 5$. The literal sharp cutoff is not replaced by this smooth trial. The signed large-conductor fourth-moment and mixed Möbius sums remain open; no improved full exponent, BT_σ, new Lean theorem, zero-location result or GRH conclusion follows.

25 Scope of the raw fourth-moment reduction

Write $c_{a,b} = \mu(a)\mu(b)\nu(ab)W(Nab/D)$ on the exact factor support in the core. Before absolute Cauchy, the normalized energy is

$$\frac{1}{D} \sum_{(a,b),(c,d) \text{ on that support}} c_{a,b}\overline{c_{c,d}} \sum_{u \in B} x_a(u)x_b(u)\overline{x_c(u)x_d(u)}.$$

Squarefreeness of ab and cd forces each positive and negative prime multiplicity to be at most one, proving $e_p \in \{0, \pm 1\}$. For the mixed term against the actual μ polynomial the exponent is $\mathbf{1}_{p|a} + \mathbf{1}_{p|b} - \mathbf{1}_{p|n}$, with the same conclusion and the same full union mask. The common possibly complex W, ν, hyperbolic support and all raw zeros stay in these identities.

In contrast, unrestricted four-column memberships $(1, 1, 0, 0)$ give $e_p = 2$, a cubic component; $(1, 1, 1, 0)$ give $e_p = 1$ although both product supports contain p. Neither the parity symmetric difference of four supports nor the symmetric difference of two nonsquarefree product supports gives the general conductor. For the interface in the core, its local character order is

j	$1, 5$	$2, 4$	3	6
$e_p = \pm 1$	6	3	2	1
$e_p = \pm 2$	3	3	1	1

Nontrivial entries have tame conductor exponent one; trivial entries remain deleted at $p \mid v$. Multiplying the interface over all six row factors, with the fixed unit/S phase, preserves every original zero, including those at sixth-power factors. All inter-factor, squarefree, ray and capacity restrictions remain external restrictions on the same row sum; this identity asserts no independence.

At a residue field of size Q, let χ have order six and ρ be nontrivial additive. The zero-extended complete sum $G_k(t) = \sum_{z \neq 0} \chi(z)^k \rho(tz)$ satisfies

	$t = 0$	$t \neq 0$
$k \not\equiv 0$ (6)	0	$\lvert G_k(t)\rvert = \sqrt{Q}$
$k \equiv 0$ (6)	$Q - 1$	-1.

For $t \neq 0$, substitution gives $G_k(t) = \overline{\chi(t)^k} G_k(1)$; for nontrivial k, expanding the norm square gives $(Q-1) - \sum_{a \neq 0,1} \chi(a)^k = Q$. The proof applies at split and inert primes. By CRT the full period-L transform vanishes if any local frequency on Q_j is zero; otherwise its magnitude is

$$(NQ_j)^{1/2} \prod_{p \mid Z_j} \left| -1 + Np\, \mathbf{1}_{\text{local frequency at } p \text{ is } 0} \right|.$$

Canceled common primes thus supply Ramanujan factors.

Exact deletion, with all remaining restrictions kept in F, gives

$$\sum_r \theta(r) F(r) \mathbf{1}_{(r,Z)=1} = \sum_{d \mid Z} \mu(d) \theta(d) \sum_z \theta(z) F(dz).$$

For a squarefree pivot let E be the radical of its fixed excluded primes, including Z and the relevant other row factors, with conductor primes omitted since θ already vanishes there. The additional squarefree identity is

$$\mu(r)^2 \mathbf{1}_{(r,E)=1} = \sum_{\substack{h^2 \mid r \\ (h,E)=1}} \mu(h) \sum_{\substack{d \mid E \\ d \mid r}} \mu(d).$$

Since $(d,h) = 1$, its lifted row sum has terms $\mu(d)\mu(h)\theta(d)\theta(h)^2 \sum_z \theta(z) F(dh^2 z)$. The scaled F retains the exact window, primary/ray restrictions and other masks. Divisor triangles cost at most $(HD)^\epsilon$; they do not restore a forbidden nonzero.

The previous fixed-radial-profile high-gcd proof (`research/short-row-high-gcd.md`, independently reviewed in `planning/reviews/book/short-row-formal.md`) already bounds its off-diagonal tail by $O_{\Phi,\epsilon}(D^{2+\epsilon}/G_0^2)$ for $Nh \geq G_0$. For annular $m = hr, n = ht$, $Nq \leq Y$ implies $Nh \geq D/\sqrt{Y}$, so the old smooth-scope bound is $O_{\Phi,\epsilon}(D^\epsilon Y)$. The present elementary count $O(D\sqrt{Y})$, factor multiplicities $r_U(m) r_U(n) \ll_\epsilon D^\epsilon$, and $\#B_A = O(R_0)$ give only $O(\|W\|_\infty^2 D^\epsilon R_0 \sqrt{Y})$ on a literal sharp block. This is a weaker conductor-form recovery in its own bookkeeping scope, not a new high-gcd theorem; the smooth estimate is not silently transferred to a selected sharp off-diagonal piece. The full A_3 face and neighborhood slack are also inherited.

Finally, primitive smooth Poisson completion has bound $O_V(\min(A_j, \sqrt{Nq}))$ on the original support. For $Nq > P^2$ and $1 \leq j \leq 5$, every scaled length $A_j/(Nd\,Nh^2)$ is below $\sqrt{Nq}$, so even this favorable trial takes its trivial length term. Its constant uses fixed smooth seminorms and need not stay bounded when a sharp boundary is approached. In the admissible subsector $(m,n) = 1$, $Nq = Nm\,Nn \asymp D^2$. A complete zero mean does not control the nonzero frequencies of the short original masked row window. The unresolved target is their joint signed sum with weights $r_U(m) r_U(n) \lambda_m \overline{\lambda_n}$ and $r_U(m) \lambda_m \overline{\lambda_n}$, before per-conductor absolute estimates. No new global analytic conclusion is claimed.

Classical reference NIST Digital Library of Mathematical Functions, Section25.15, Dirichlet L-functions, equations 25.15.5–6; primitive functional equation and finite Gauss sums (snapshot pinned 4 November 2026). The normalization and explicit finite Gauss sums used here are stated above.